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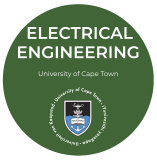
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Fri 23 October, 2026

Abstract

Thesis Title

Joab Gray **Kloppers**

Fri 23 October, 2026

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List of Symbols

Symbol	Description	Unit

Chapter 1

Introduction

1.1 Background

1.1.1 Sand prawns and bioturbation

Bioturbation, the reworking of sediment particles and the ventilation of burrows by animals [27], is one of the main ways in which benthic fauna shape soft-sediment ecosystems. Burrowing thalassinidean shrimps, the group to which sand prawns belong, are among the most influential bioturbators. Pillay and Branch [38] describe how their burrowing brings buried sediment to the surface, builds mounds that change the structure and increase the area of the sediment surface, and alters the physical and chemical properties of the sediment through burrow construction and irrigation.

On the South African coast, the sand prawn *Callichirus kraussi* is a key ecosystem engineer. Venter et al. [50] showed experimentally that sand prawns filter phytoplankton from the overlying water by pumping water in both directions through their burrows. Long-term estuarine data further showed reductions in phytoplankton biomass of roughly 70% in sand prawn-dominated areas, which suggests that sand prawns help to buffer coastal ecosystems against eutrophication. Their methodology, however, relied on manual visual counts of burrow openings, which are laborious and describe only one aspect of the altered surface. This project forms part of a wider study of sand prawn bioturbation on the intertidal flats of the Langebaan Lagoon. [TODO: Name the overarching project and research group, and cite or acknowledge it.]

1.1.2 Measuring sediment microtopography

Bioturbators leave mounds, burrow openings and depressions at the sediment surface. The microtopography of the surface, its relief at millimetre to centimetre scales, therefore carries a signature of the activity beneath it. Schenone et al. showed in laboratory incubations that sediment microtopography predicted localised benthic ecosystem functioning [43]. In a later field study using structure-from-motion (SfM) photogrammetry, they found that it explained a substantial proportion of the variance in biodiversity and ecosystem functioning across subtidal habitats [44]. Microtopography is summarised by areal roughness parameters. Of these, rugosity is the ratio of the true surface area to its projected area, and is closely related to the developed interfacial area ratio defined in ISO 25178-2 [2].

Measuring microtopography in the field is not straightforward. Traditional sampling methods are invasive and damage the fragile structures being measured. Non-contact alternatives such as LiDAR and SfM photogrammetry can be expensive and cumbersome to process [6]. Azhar et al. [6] addressed this with a low-cost framework. A consumer RGB-D camera, the Asus Xtion Pro Live, captured depth images of intertidal sediment, from which roughness parameters were computed after detrending. Low-cost RGB-D sensors have similarly been used to quantify habitat complexity [24]. The Xtion relies on structured light, which is degraded by strong ambient light [19]. It therefore had to be operated inside an opaque enclosure, which limited the area of each scan.

1.2 Problem statement

The overarching project adapted the framework of Azhar et al. [6] to an Intel RealSense D456, mounted on a portable aluminium rig above the sediment. The D456 is an active stereo camera [3]. It computes depth on the camera by matching texture between two infrared imagers, and a projected infrared dot pattern adds texture where the surface has little of its own [15, 25]. Unlike a structured-light camera, it can therefore still recover depth from naturally textured surfaces when ambient light washes out the projected pattern. As inherited, however, the system had three shortcomings.

First, its configuration had not been systematically evaluated. This covers the working distance, viewing angle, lighting, stream resolution, depth unit, visual preset and post-processing filters. Published evaluations of depth cameras [9, 19, 40] consider other camera models and mostly indoor scenes of planar or manufactured objects. None characterises the D456 on sediment-like relief at the sub-metre working distances of the rig.

Second, each surface was scanned at least four times from different mount rotations, but each scan was analysed individually. A single scan is noisy and has viewpoint-dependent errors. It also contains holes wherever relief hides the surface from the camera. The repeated scans were not combined to reduce these effects.

Third, the accuracy of the reconstructions, and of the roughness parameters computed from them, had not been quantified against a known ground truth. [TODO: Confirm that no ground-truth validation was done before this project.]

This project therefore characterises the D456 for close-range scanning of sediment-like surfaces. It also develops a processing pipeline that fuses multi-view scans into a single reconstruction, from which roughness parameters are computed and validated.

1.3 Project objectives

The objectives of this project were to:

- Characterise the D456 as functions of working distance, camera tilt and ambient lighting, using planar targets.
- Characterise the D456 stream settings (resolution, depth unit and visual preset) and post-processing filter settings, by evaluating scans of 3D-printed reference objects of known geometry across the different parameters.
- Develop a pipeline that automatically crops each scan to the analysis region marked by AprilTags, registers scans taken at different tilts and rotations, and fuses them into a single surface. Characterise how reconstruction accuracy depends on the combination of views.
- Develop a roughness estimation pipeline that levels, detrends and grids the fused surface and computes standardised areal roughness parameters, including rugosity. Validate it against the reference objects and against laboratory sand.
- Investigate the automated detection and counting of discrete biogenic features, such as burrow openings and mounds, in the fused surfaces.

1.4 Scope, limitations and assumptions

Scope. The project was carried out over 12 weeks, with final submission on 27 October 2026. It was limited to the existing hardware, and no other cameras were evaluated. The hardware comprised:

- A single Intel RealSense D456 on a 1 m aluminium-extrusion frame with a height-adjustable cross member.
- A manually rotated camera mount, turned in 45° increments, on a servo-driven tilt stage.
- Four adjustable AprilTag boundary markers.
- An overhead shading tarp.

Testing used planar targets, three 3D-printed reference objects, and sand collected from the Langebaan Lagoon. The sand was scanned in the laboratory in dry, wet and waterlogged states. Field deployment on the intertidal flats was not possible within the time available.

Limitations. The characterisation and the chosen configuration target smooth, gently undulating relief of a few centimetres, representative of sand prawn mounds. They may not transfer to objects with sharp edges. The fused surface is represented as a single-valued height field, so overhangs such as burrow walls cannot be represented. It is also smoothed with a 10 mm moving least squares radius by default, so relief at shorter wavelengths is attenuated. Each characterisation condition was recorded once, so the repeatability of the set-up was not quantified. Automated feature detection was not implemented within the project timeframe and is discussed as further work in chapter 10. [TODO: Update this sentence if feature detection is implemented before submission.]

Assumptions. The following assumptions were made:

- The factory calibration of the D456 (camera intrinsics and depth-to-colour extrinsics) is accurate, and the camera was not recalibrated. [TODO: Confirm that no recalibration was performed.]
- The surface does not change during a capture session, and the AprilTags remain fixed relative to it, so every scan in a session observes the same region.
- The sediment within the analysis region is approximately planar at the scale of the region, with relief superimposed on it.
- The camera is stationary while each scan is captured, so its accelerometer measures gravity alone.
- The 3D-printed reference objects match their CAD models closely enough to serve as ground truth. These reference objects were printed using a Bambu lab X1 carbon.

1.5 Plan of development

Chapter 2 reviews the literature on sand prawn bioturbation, microtopography as a proxy for benthic activity, and depth sensing for surface measurement. Chapter 3 presents the theory of stereo depth capture, registration, fusion, evaluation and roughness estimation. Chapter 4 defines the system requirements and describes the rig and software architecture. Chapter 5 characterises the D456 camera across different conditions. Chapter 6 describes the cropping, fusion and roughness estimation pipeline. Chapter 7 validates the complete pipeline on reference objects and laboratory sand. Chapter 8 provides a summary of the results found in the project, chapter 9 draws conclusions against the objectives, and chapter 10 outlines further work.

Chapter 2

Literature review

This chapter aims to provide a review on the literature surrounding the project, and the motivations behind it, as well as identifying the contribution towards the literature that can be made. This review will first explore why sand prawn bioturbation matters, and then explain how surface microtopography, and surface roughness, is a valid proxy for bioturbation. The review will then explore techniques that have been used to measure sediment. Next a review on how cameras have been characterised in existing literature, and how depth scans have been combined. Finally, a research gap is addressed, and explains the purpose of this project.

2.1 Sand prawn bioturbation and its ecological significance

Kristensen et al. [27] argue that bioturbation needs a precise definition that separates its components, mainly the reworking of sediment particles and the ventilation of burrows, so that each can be measured and compared across studies. Burrowing thalassinidean shrimps are among the most intense bioturbators. In their review, Pillay and Branch [38] describe them as ecosystem engineers. By excavating burrows and ejecting sediment at the surface they alter grain size, organic content, porewater chemistry and surface topography, and these changes propagate to other fauna and flora. The effects of this activity on the surface is predominantly in the form of mounds, depressions and burrow openings. The review reports mounds of about 5.4 cm in height and 11.7 cm in width for *Callianassa subterranea* in the North Sea, and of about 3 cm in height and 15 cm in width for *Pestarella candida* in the Mediterranean. These values indicate the scale of relief a measurement system must resolve, centimetre-scale heights over decimetre-scale wavelengths.

On the South African coast the dominant species is the sand prawn *Callichirus kraussi*, named *Callianassa kraussi* in the older literature. In Langebaan Lagoon, Nel and Branch [32] found that it occupied most of the intertidal area and dominated its sandy sediments, and that its distribution followed sediment characteristics. Its engineering effects are substantial. Branch and Pringle [8] reported sediment turnover of the order of 12 kg of wet sediment per square metre per day, which buried benthic microalgae and altered bacterial and meiofaunal communities. In caging experiments, Pillay et al. [37] found that excluding sand prawns increased the abundance of surface grazers and of both suspension-feeding and deposit-feeding polychaetes and bivalves, while burrowing infauna were unaffected. Venter et al. [50] extended the role of the species to the water column. Bi-directional pumping through the burrows, with adsorption of phytoplankton onto the burrow walls, reduced phytoplankton biomass by roughly 50 % in experiments, and long-term estuarine data showed reductions of roughly 70 % in prawn-dominated areas, which suggests that sand prawns help to buffer coastal ecosystems against eutrophication.

Because these effects scale with prawn density and activity, understanding them requires the activity to be measured. The usual approaches are sediment cores, which are destructive and slow to process, and counts of burrow openings, which are laborious and capture only one kind of surface feature. Surface structure is also ecologically important in its own right: Thrush et al. [48] showed that disturbances which homogenise surficial sediments and reduce habitat structure can have important consequences for soft-sediment biodiversity. The relief of the sediment surface is therefore both a product of bioturbation and a habitat variable, which raises the question of whether it can be measured in place of the animals themselves.

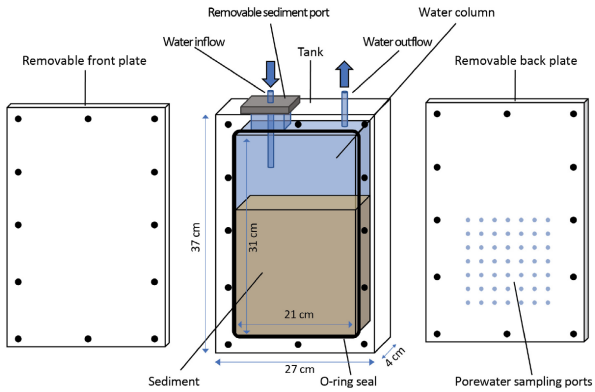


Figure 2.1: Custom incubation and flow through chamber by Schenone et al [43]

2.2 Sediment microtopography as a proxy for benthic activity

The case for microtopography as a proxy rests mainly on a sequence of studies by Schenone and colleagues. In a multi-site field study, Schenone and Thrush [42] found that the biogenic features created by infaunal species at the sediment surface predicted ecosystem functioning better than the density of those species, and identified this as particularly suited to large-scale mapping and monitoring. Schenone et al. [43] then tested the link under controlled conditions in custom flow-through, gas-tight tanks containing a tellinid bivalve and a maldanid polychaete, seen in Figure 2.1. Rugosity explained between 35 and 61 % of the variability in oxygen, ammonium and nitrate plus nitrite fluxes. Their rugosity was obtained from the sediment profile imaged through the tank wall, a single two-dimensional slice of the surface. Schenone et al. [44] took the approach into the field, reconstructing the seafloor of several subtidal habitats with structure-from-motion (SfM) photogrammetry, and found that microtopography consistently explained more than half of the variance in environmental variables, benthic fluxes, biodiversity metrics and functional traits.

Two caveats limit how directly these results transfer to this project. The studies were subtidal, in New Zealand, and involved different species, so the relationship has not been demonstrated for *C. kraussi* on exposed intertidal flats. The relationships are also correlative as they establish microtopography as a surrogate rather than as a mechanism. Testing the ecological relationship at Langebaan is the purpose of the wider study to which this project contributes. The task here is to provide measurements accurate enough for that test.

How roughness is quantified matters as much as whether it is measured. Benthic ecology traditionally uses chain-and-tape rugosity, the ratio of the length of a chain laid along the contours of the bed to the straight line distance between its ends. Friedman et al. [13] note that this samples a single profile and tends to overestimate rugosity when the chain wanders sideways. They instead computed rugosity from stereo-derived meshes as surface area relative to the area projected onto a plane of best fit, which decouples rugosity from slope and allows it to be evaluated over windows of different sizes. Storlazzi et al. [47] derived rugosity from SfM reconstructions of existing reef video and found it more effective and quantitatively powerful than chain-and-tape. Surface metrology provides a standardised vocabulary for the same quantities. ISO 25178-2 defines areal height, spatial and hybrid parameters on a scale-limited surface [2, 28], of which the developed interfacial area ratio S_{dr} is rugosity expressed as a percentage excess area (Chapter 3). Azhar et al. [6] adopted a comparable set, computing nine morphological characteristics, including arithmetic mean roughness, mean peak height and rugosity, from detrended depth images.

These parameters are not independent of how the surface is sampled. Jacobs et al. [23] show that common roughness parameters depend on the scan size and pixel resolution chosen by the researcher, and recommend the power spectral density as a description that is not biased by these choices. Slope-based parameters such as S_{dq}, S_{dr} and rugosity are dominated by the shortest resolved wavelengths, where sensor noise and smoothing also act (Chapter 3). Rugosity from a depth camera is therefore only interpretable when the noise floor and effective resolution of the measurement are known. Without that, a change in rugosity cannot be attributed to

the sediment rather than to the instrument.

2.3 Non-contact measurement of sediment surfaces

A system for this application must resolve millimetre-scale relief over patches of a few tenths of a square metre, be portable and fast enough to use within a low-tide window, leave the surface undisturbed, and operate in daylight on dry, wet and waterlogged sand. Coring and contact profiling fail the third requirement because they damage the fragile structures being measured [6]. The non-contact options differ mainly in cost, processing effort and tolerance of sunlight.

Terrestrial laser scanning (TLS) is accurate and is used on sandy coasts, where its return intensity has even been calibrated against surface moisture [35], but the instruments are expensive and the data cumbersome to process [6]. SfM photogrammetry is inexpensive in hardware and can reach spatial resolutions orders of magnitude finer than large-scale lidar or sonar surveys [47], which is why it underpins the field work of Schenone et al. [44]. It does, however, require many overlapping images, an external scale reference and offline processing, so the result is not available while still on site.

Consumer depth cameras offer immediate, metrically scaled depth at low cost. Mankoff and Russo [31] introduced the structured-light Kinect to the Earth sciences, reporting depth resolution of 1 mm or better at 0.5 m, degrading to about 75 mm at 5 m, and demonstrating it in glaciology, stream bathymetry and geomorphology. Kamal et al. [24] used the same sensor to quantify the three-dimensional complexity of mangrove roots. Azhar et al. [6], the framework on which this project builds, used an Asus Xtion Pro Live inside a semi-opaque container to image 0.3 m^2 patches of intertidal sediment at seven sites (82 surfaces). They first tested four variants of a geometric detrending model on simulated surfaces, then applied the two best-performing models, linear and quadratic, to the field data, and found strong agreement between the two in the extracted surface measures. The container was a necessity rather than a choice. A structured-light camera triangulates against its own projected pattern, which sunlight washes out, so shading fixed both the size of each scan and its single viewpoint.

Sunlight tolerance therefore decides the choice of sensor. Structured-light and time-of-flight (ToF) cameras rely entirely on their own near-infrared illumination. An active stereo camera instead matches texture between two infrared imagers, and its projected dot pattern only supplements the scene texture. On a naturally textured surface in strong light it degrades towards a passive stereo camera rather than failing outright [15, 25]. Vit and Shani [51] tested four sensors outdoors at several distances and lighting conditions: a ToF Kinect v2, a structured-light Orbbec Astra S, a coded-light Intel SR300 and an active stereo Intel D435. They concluded that the D435 was the most viable for close-range field measurements. Purely passive stereo cameras such as the ZED 2 also tolerate sunlight but require a GPU-equipped host to compute depth [40], whereas the RealSense D400 cameras compute depth on board. The D456 used here is the IP65-rated variant of the D455 [3], which also suits wet intertidal work. Active stereo is thus the best-supported low-cost option, but the evidence for it comes from other camera models and other surfaces.

2.4 Characterisation of active stereo depth cameras

The manufacturer literature describes how the RealSense cameras work and how to configure them. Keselman et al. [25] cover the R200 and D400 (RS400) designs, their correlation algorithm and models of expected performance, including a minimum range set by the disparity search range, a maximum range limited by signal-to-noise ratio, and artefacts from laser speckle. The tuning guide of Grunner-Jepsen et al. [16] gives recommendations on resolution, visual presets, depth units and laser power. Grunnet-Jepsen and Tong [17] describe the decimation, edge-preserving spatial, temporal and hole-filling filters and a recommended order for them. Grunner-Jepsen et al. [15] show that, because the D400 cameras do not need a known pattern, an external projector can add texture beyond that of the internal one. Intel's test methodology [1] defines the plane-fit metrics used in this project's planar tests. These documents are general-purpose guidance, none evaluates the

recommended settings against ground-truth relief at the millimetre scale relevant to sediment.

Independent metrological studies exist for related models. Carfagni et al. [9] characterised the D415 between 100 and 1000 mm, combining the VDI/VDE 2634 Part 2 guidelines with strategies from the literature. Servi et al. [46] applied ISO 10360-13 to the D415, D455 and L515 between 100 and 1500 mm, adding flat-object and 3D-reconstruction tests. The D415 gave the best reconstructions at short range, the L515 the smallest systematic depth errors and the D455 the best results in the standardised tests. These are the closest precedents for this work, since the D456 shares the optical module of the D455 [3], and their combination of planar and three-dimensional artefacts is mirrored in the planar and reference-object tests of this project. Both, however, were laboratory studies of manufactured artefacts under controlled lighting.

Multi-sensor comparisons broaden the picture but share the same limitation. Halmetschlager-Funek et al. [19] evaluated ten structured-light, active stereo and ToF sensors for bias, precision, lateral noise, sensitivity to materials and lighting, and interference between multiple sensors. Their tests were indoors only and achieved by only artificial light sources. Rustler et al. [40] compared four stereo cameras on a planar surface, a plastic doll and household objects, recording over 3000 frames per camera. The D435 performed best within 1 m, with errors below 1 cm in all scenarios, while the D455, ZED 2 and OAK-D Pro were better at longer range. This ranking cannot simply be transferred to the D456. The D455 optical module has a 95 mm baseline against the 50 mm of the D435 [3], which lowers far-range noise but raises the minimum working distance, so results for the D435 do not carry over. Only Servi et al. [46] cover the D455 module at the 0.6 to 0.9 m range of the rig, and only in the laboratory. Table 2.1 summarises these studies.

Table 2.1: Published depth-camera characterisations relevant to this project. SL: structured light; AS: active stereo; PS: passive stereo; ToF: time of flight.

Study	Sensors (technology)	Targets	Environment
Halmetschlager-Funek et al. [19]	Ten sensors (SL, AS, ToF)	Planar targets, several materials	Indoor
Carfagni et al. [9]	D415 (AS)	VDI/VDE 2634 artefacts, 0.1–1 m	Laboratory
Servi et al. [46]	D415, D455 (AS); L515 (ToF lidar)	ISO 10360-13 artefacts, flat and 3D objects, 0.1–1.5 m	Laboratory
Rustler et al. [40]	D435, D455, OAK-D Pro (AS); ZED 2 (PS)	Plane, doll, household objects, up to 4 m	Indoor
Vit and Shani [51]	Kinect v2 (ToF), Astra S (SL), SR300 (SL), D435 (AS)	Maize plants, close range	Outdoor, varied light
Azhar et al. [6]	Xtion Pro Live (SL)	Plane; intertidal sediment	Outdoor, shaded
This work	D456 (AS)	Planes, 3D-printed relief, sand, 0.6–0.93 m	Indoor, outdoor

Noise models offer a route from characterisation to better processing. Nguyen et al. [34] derived lateral noise of the structured-light Kinect from the spread of depth values across depth edges, which reflects the point spread function of the optics. They derived axial noise from residuals about planes fitted at varying distances and angles. Feeding the resulting model into KinectFusion recovered finer detail and improved pose estimation. The method itself is not specific to the Kinect, and the distance and angle sweeps of the project objectives follow the same logic for the D456.

Wet and waterlogged sand presents a further, largely untested, condition. No study was found that characterises a RealSense camera, or any active stereo camera, on wet or waterlogged sediment. The optics literature indicates why it matters. Nolet et al. [35] measured the reflectance of beach sand from 350 to 2500 nm and found that reflectance increased non-linearly as the sand dried, across the whole range. Wet sand is darker in the near-infrared as well as the visible, which can be visualised in Figure 2.2. It can be inferred that wet sediment returns less of the projected pattern and offers lower image contrast for matching. Standing water adds specular reflection and refraction at its surface. Mankoff and Russo [31] used an active depth camera for stream bathymetry, where the water surface refracts the measurement. Halmetschlager-Funek et al. [19] likewise found that some sensors failed on reflective materials under bright light. The behaviour of the D456 on dry, wet and waterlogged sand therefore has to be established experimentally.

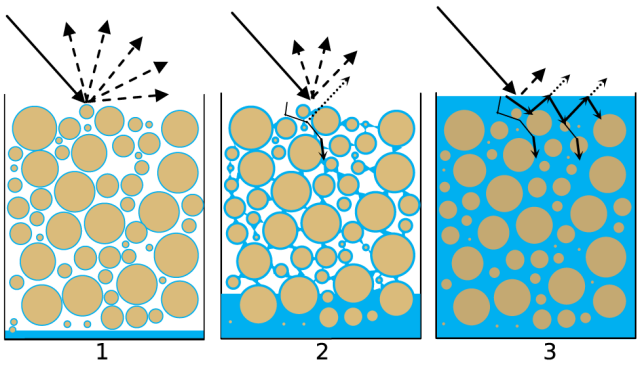


Figure 2.2: Conceptual representation of the non-linear decrease of spectral reflectance as moisture content in the sand matrix increases. Nolet et al. [35]

2.5 Registration and fusion of multi-view depth scans

Combining depth scans from several viewpoints is well established in computer vision. Curless and Levoy [11] integrated aligned range images into a cumulative weighted signed distance function, which averages noise, fills gaps and weights measurements by their reliability. KinectFusion [33] applied the same representation in real time to a moving consumer depth camera, tracking each frame against the growing model with ICP, and showed that the fused model contains detail absent from any single frame. Nguyen et al. [34] improved on this by weighting measurements with a sensor noise model. These methods were developed for objects and rooms, and their quality has been judged by geometric accuracy rather than by the roughness parameters derived from the result.

Two further results bear on sediment. First, nearly flat surfaces are a degenerate case for registration. Gelfand et al. [14] show that ICP cannot constrain sliding within a plane or rotation about its normal, so a near-planar bed needs external constraints such as fiducial markers, for which AprilTags [36, 52] are a robust choice, or inertial measurements. Second, in benthic mapping, fusion is normally done at survey scale. Friedman et al. [13] built rugosity maps from stereo imagery registered with vehicle navigation, rather than by fusing repeated close-range views of a single patch. Azhar et al. [6] analysed each surface from a single fixed viewpoint inside the container, and the inherited system for this project likewise analysed each of its repeated scans individually. No study was found that fuses multi-view depth scans of a sediment patch and quantifies how the choice of views affects the accuracy of the reconstruction and of the roughness parameters derived from it.

2.6 Summary and research gap

The literature establishes that *Callichirus kraussi* is a dominant ecosystem engineer of the Langebaan intertidal, whose activity reshapes the sediment surface and affects benthic and pelagic processes. Sediment microtopography, and rugosity in particular, has explained a substantial share of the variability in benthic fluxes and biodiversity in subtidal systems, which makes it a credible proxy worth measuring. Roughness parameters, however, depend on the scale and noise of the measurement. Low-cost depth cameras can capture millimetre-scale relief in the field. Structured light needs shading, which restricted the framework of Azhar et al. [6] to a small, single-view patch, whereas active stereo is the best-supported alternative for outdoor use. Against this background, five gaps remain, each addressed by one objective in section 1.3:

- The D455 optical module shared by the D456 has been characterised only in the laboratory, on manufactured artefacts [46]. Its noise, bias and fill rate at the working distances, tilts and outdoor lighting of the rig, and on sediment-like surfaces, are unknown, and results for other models do not transfer across baselines.
- Manufacturer guidance on resolution, depth unit, preset and post-processing filters has not been validated

against ground-truth relief of the shape and scale of sand prawn mounds .

- Multi-view fusion has not been applied to close-range scans of a sediment patch, and the effect of view combination on reconstruction accuracy is unknown. Near-flat beds also defeat purely geometric registration.
- Rugosity and related parameters from RGB-D data have not been validated against surfaces of known geometry, despite their sensitivity to noise and resolution.

Chapter 3

Theoretical Background

This chapter outlines the theory behind the project, seperated by each stage: depth measurement, registration, fusion, evaluation and roughness estimation. The main aim of this section is to provide a high level description of some of the fundamentals. Implementation details and parameter choices are given in chapter 4 and chapter 6.

3.1 Depth Capture

Seeing Double Vision Can Be Good

The Intel RealSense D456 is an IP65-rated version of the D455, built from the same D450 optical module and D4 vision processor [3]. It contains two global-shutter imagers separated by a baseline of 95 mm, an infrared (IR) laser projector, a global-shutter RGB camera and an inertial measurement unit (IMU). The Camera features the D4 processor, which computes depth on the camera, so the host receives depth images rather than stereo image pairs.

3.1.1 Camera Model

Each imager is modelled as a pinhole camera [20]. A point $\mathbf{X}_w$, in a world frame is mapped to the pixel (u, v) by

$$\lambda\,(u,\,v,\,1)^\top = \mathbf{K}\,[\mathbf{R}\mid\mathbf{t}]\,(\mathbf{X}_w^\top,\,1)^\top, \qquad \mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{3.1}$$

where $\mathbf{X}_w = (X_w, Y_w, Z_w)^\top$ is the point in the world frame, and $(X, Y, Z)^\top = \mathbf{R}\mathbf{X}_w + \mathbf{t}$ is the same point in the camera frame, whose depth Z equals λ. The intrinsic matrix $\mathbf{K}$ holds the focal lengths f_x,f_y in pixels and the principal point (c_x, c_y). The extrinsic parameters $\mathbf{R}$ and $\mathbf{t}$ give the pose of the camera relative to another frame, such as the world or a second sensor. Real lenses add radial and tangential distortion, and calibration estimates all of these parameters, typically from images of a known planar target [53]. The D456 is factory calibrated, and because its stereo images are rectified on the camera, the depth stream is described by a distortion-free pinhole model while the colour stream has its own distortion coefficients.

Inverting Equation 3.1 for a pixel with measured depth Z gives the corresponding point in the camera frame,

$$X = \frac{(u - c_x)\,Z}{f_x}, \qquad Y = \frac{(v - c_y)\,Z}{f_y}, \tag{3.2}$$

which is how a depth image becomes a point cloud. Depth is aligned to the colour image by back-projecting each pixel, transforming it with the depth-to-colour extrinsics and projecting it with the colour intrinsics.

3.1.2 Stereo Triangulation

After rectification the image rows of the two cameras are aligned, so a scene point appears on the same row in both images and the search for its match is one-dimensional [20]. In the geometry of Figure 3.1, a point at depth Z projects to horizontal image coordinates x_L and x_R, measured from each principal point, and similar triangles give

$$Z = \frac{fB}{d}, \qquad d = x_L - x_R, \tag{3.3}$$

where f is the focal length in pixels, B the baseline and d the disparity. The camera finds d for each pixel by comparing small image patches along the row, and interpolates the best match to a fraction of a pixel [25].

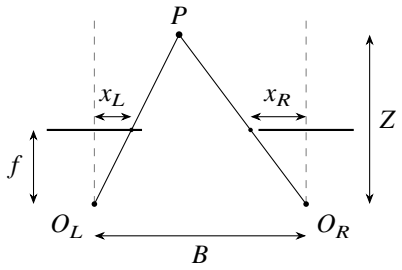


Figure 3.1: Rectified stereo geometry: camera centres O_L, O_R, baseline B, focal length f and the projections x_L, x_R of a point P at depth Z.

Differentiating Equation 3.3 gives the depth uncertainty caused by an uncertainty σ_d in disparity,

$$\sigma_Z = \frac{Z^2}{fB}\,\sigma_d. \tag{3.4}$$

For a fixed matching error, depth noise grows with the square of the range and falls with a longer baseline or focal length [25]. Since f in pixels is proportional to the horizontal resolution, a lower resolution also increases the noise. A constant σ_d is optimistic: it depends on the contrast of the matched texture, which falls as the projected pattern dims with distance or is washed out by ambient light.

The disparity search also has a finite range. The D4 processor searches 126 disparities [18], so the minimum measurable depth is $Z_{\min} = \frac{fB}{126}$. At 1280×720, $f \approx \frac{640}{\tan(87°/2)} \approx 674\,\text{px}$, giving $Z_{\min} \approx 0.51\,\text{m}$, consistent with the specified 0.52 m [3]. Equation 3.4 and $Z_{\min}$ together set the working-distance trade-off characterised in chapter 5. the camera should be as close as $Z_{\min}$ and the required field of view allow.

3.1.3 Passive Stereo, Structured Light and Active Stereo

Passive stereo matches natural texture between two cameras, and fails where the surface is uniform because many patches along a row look alike. Structured light, used by the Asus Xtion in the system of [6], projects a known pattern and triangulates between the projector and a single camera, so the pattern itself must be known and calibrated. Active stereo, the principle of the D456, combines a stereo pair with a projector of a semi-random dot pattern, which can be seen in chapter B, [15, 25]. Matching is still performed between the two cameras, so the pattern does not need to be know, the main purpose is to add aditional texture onto the surface. On a featureless surface, such as the white board of chapter 5, depth therefore depends entirely on the dots and fails when sunlight, whose near-infrared component can outshine them, washes them out. On a naturally textured surface in sunlight the camera behaves as a passive stereo camera and still returns dense depth. Both behaviours were observed in chapter 5. The RGB camera has an IR-cut filter, so the dots do not appear in the colour image.

3.2 Registration

We Need To Align Ourselves

Each scan is expressed in the frame of the camera at the moment of capture, so the same point on the sediment has different coordinates in scans taken at different rotations and tilts. Registration finds the rigid transformation that maps one scan into the frame of another, and is needed both for fusion (section 6.2) and for comparison with a reference model (section 4.5).

3.2.1 Rigid Transformations and Closed-Form Alignment

A rigid transformation maps a point $\mathbf{p}$ to $\mathbf{R}\mathbf{p} + \mathbf{t}$, where $\mathbf{R}$ is a rotation matrix ($\mathbf{R}^\top\mathbf{R} = \mathbf{I}$, $\det\mathbf{R} = +1$) and $\mathbf{t}$ a translation, giving six degrees of freedom. If N corresponding point pairs $(\mathbf{p}_i, \mathbf{q}_i)$ are known, the transformation that minimises $\sum_i \|\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|^2$ has a closed-form solution [5, 49]. With centroids $\bar{\mathbf{p}}$ and $\bar{\mathbf{q}}$, and the singular value decomposition $\mathbf{H} = \sum_i (\mathbf{p}_i - \bar{\mathbf{p}})(\mathbf{q}_i - \bar{\mathbf{q}})^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$,

$$\mathbf{R} = \mathbf{V}\,\mathrm{diag}(1, 1, \det(\mathbf{V}\mathbf{U}^\top))\,\mathbf{U}^\top, \qquad \mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}. \tag{3.5}$$

The determinant term ensures a proper rotation rather than a reflection when the data are noisy [49]. At least three non-collinear pairs are needed because if the points lie close to a line, rotation about that line is poorly determined, which is why the tag-based alignment in section 6.2 takes the tilt from the IMU when the visible tags are nearly collinear. A stationary accelerometer measures only gravity, which fixes the two tilt angles but not the rotation about the vertical axis, which can be determined from tag identification or surface geometry.

3.2.2 Fiducial Markers

AprilTags are square black-and-white markers that encode an identifier in a binary grid. The codes are chosen to keep a large minimum Hamming distance between all identifiers in all four rotations, so a tag is rarely misidentified even when some bits are read incorrectly [36]. The detector fits quadrilaterals to the boundaries between light and dark regions, decodes the bits inside each candidate, and returns each tag's identifier and the sub-pixel positions of its four corners [52].

The corners of a planar tag of known size are related to their image positions by a homography $\mathbf{H} \simeq \mathbf{K}[\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}]$, where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the first two columns of the tag's rotation matrix [20, 53]. Given $\mathbf{K}$, the pose follows from $\mathbf{K}^{-1}\mathbf{H}$ with $\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$. This monocular estimate rests on four closely spaced points and is sensitive to corner noise. For a small or nearly fronto-parallel tag, two different orientations can fit the corners almost equally well [45]. The pipeline instead uses the depth data, intersecting the corner pixel rays with a plane fitted to the measured depth of each tag (section 6.1). This gives 3D corners in every scan, so aligning two scans becomes a 3D-to-3D problem solved with Equation 3.5.

3.2.3 Feature-Based Coarse Alignment

When no prior pose is available, correspondences must be found from the shape of the surface. Fast Point Feature Histograms (FPFH) describe the geometry around each point by the angles between its surface normal and those of its neighbours, expressed in a local coordinate frame. Three angular features are each binned into 11 bins and concatenated into a 33-element histogram [41]. Because only relative angles are used, the descriptor does not change under rigid motion. Candidate correspondences are formed by pairing points with similar descriptors, and many of these pairs are wrong.

Random Sample Consensus (RANSAC) estimates the transformation in the presence of these outliers [12]. It repeatedly selects three correspondences at random, computes their transformation with Equation 3.5, and counts the inliers (correspondences brought within a distance threshold), keeping the transformation with the most inliers. If a fraction w of the correspondences is correct, drawing at least one all-inlier sample with probability p takes $N = \log(1-p)\,/\,\log(1-w^3)$ trials, or about 4600 for $w = 0.1$ and $p = 0.99$. On flat or smoothly undulating

sediment most points have nearly identical descriptors, so w is small and an incorrect transformation can gather as many inliers as the correct one. This is why feature matching is only a fallback in this pipeline (section 6.2).

3.2.4 Iterative Closest Point

Iterative Closest Point (ICP) refines an approximate alignment by alternately pairing each source point with its nearest target point and solving for the transformation that minimises the distance between the pairs. The point-to-point form [7] minimises Euclidean distance and is solved with Equation 3.5. The point-to-plane form [10] minimises the distance from each source point to the tangent plane at its paired target point,

$$E(\mathbf{R}, \mathbf{t}) = \sum_i \Big((\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i) \cdot \mathbf{n}_i \Big)^2, \tag{3.6}$$

where $\mathbf{n}_i$ is the target normal. This has no closed-form solution, but for the small rotations of a refinement step it can be linearised and solved as a six-parameter linear least-squares problem [30].

Two scans never sample exactly the same points. Point-to-point ICP penalises the resulting tangential offsets, which slows convergence, whereas point-to-plane ICP lets points slide along the surface and on smooth surfaces usually converges in far fewer iterations and to a more accurate pose, at the cost of requiring normals [39]. Both forms are local, descending to the nearest minimum of the cost, so the initial pose must already be close, and both are limited by the geometry of the surface rather than by the error metric. On a plane, translation within the plane and rotation about its normal leave the point-to-plane cost unchanged and change the point-to-point cost only slightly, so these motions are effectively unconstrained [14]. A nearly flat sediment bed is close to this degenerate case, so its in-plane pose must be fixed by the tags or the coarse stages, with ICP refining the remaining degrees of freedom.

3.3 Fusion

Putting Bits Together

[TODO: I am not liking this fusion section]

Volumetric methods store a weighted running average of the signed distance to the surface in a voxel grid and extract the surface where it is zero [11, 33]. For a nearly flat bed, a 2.5D height field $z(x, y)$ on a grid aligned with the mean plane is a simpler equivalent in which each cell collects the heights from all scans (section 6.2). [TODO: It cannot represent overhangs such as burrow walls]

First all scans points are weighted. When measurements have independent errors with different variances σ_k^2, the combination with the lowest variance is the weighted mean with weights $w_k \propto 1/\sigma_k^2$. Depth error grows with range (Equation 3.4) and with obliquity, therefore a surface seen at a grazing angle is foreshortened and its depth varies across each matching window, and the axial noise of depth sensors has been observed to rise sharply as the angle between the surface normal and the viewing direction approaches 90° [34]. Measurements are therefore down-weighted with both, in the spirit of the angle-based weights of [11].

A misregistered scan or stray point produces errors that are neither small nor normally distributed. One such value can move the mean arbitrarily far, while the median tolerates up to half the data being bad but discards much of the information in good data. The trimmed mean, which discards a fixed fraction of the lowest and highest values (by cumulative weight, in a weighted version) before averaging, lies between the two [21].

Moving Least Squares (MLS) fits a low-order polynomial surface to the neighbours of each point within a radius h, with weights that decrease with distance, and projects the point onto it [4, 29]. This removes noise and residual layer offsets on scales below h, but it is a low-pass filter that also attenuates relief with wavelengths comparable to or shorter than h, so h sets a lower bound on the scale of roughness the fused surface can represent.

3.4 Evaluation

Are We Winning?

[TODO: Not liking the evaluation section] Each reconstruction is first registered to the reference model (section 4.5). For each reconstructed point $\mathbf{s}_i$, d_i is its signed distance to the closest point on the reference mesh (positive outside the object), which avoids any dependence on how densely the reference is sampled.

Over N points the bias μ, standard deviation σ and root-mean-square error are

$$\mu = \frac{1}{N}\sum_i d_i, \qquad \sigma^2 = \frac{1}{N}\sum_i (d_i - \mu)^2, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i d_i^2} = \sqrt{\mu^2 + \sigma^2}. \tag{3.7}$$

The bias measures systematic offset, σ random spread, and the RMSE acts as a combination of both. The RMSE however is not the average distance, since squaring weights large errors more heavily, so the mean absolute error $\frac{1}{N}\sum_i |d_i|$ is less affected by outliers. Because registration chooses the pose that best fits the reference, part of any systematic error, such as a small scale error or tilt, is absorbed into the pose, so the metrics measure shape error after the best rigid fit rather than absolute accuracy.

Accuracy metrics are computed only over points that exist, so a reconstruction that covers only the easy parts of an object can still score well. Following [26], the precision $P(\tau)$ is the fraction of reconstructed points within a distance τ of the reference, and the completeness $R(\tau)$ is the fraction of the reference, sampled uniformly, within τ of the reconstruction. The F-score, $F = 2PR/(P + R)$, is high only when both are high.

With $d(\mathbf{a}, \mathcal{B})$ the distance from a point $\mathbf{a}$ to the nearest point of a set $\mathcal{B}$, the Chamfer distance averages nearest-neighbour distances in both directions between the reconstruction $\mathcal{S}$ and the reference $\mathcal{M}$, while the Hausdorff distance takes the worst case [22]:

$$d_{\mathrm{CD}} = \frac{1}{|\mathcal{S}|}\sum_{\mathbf{s}\in\mathcal{S}} d(\mathbf{s}, \mathcal{M}) + \frac{1}{|\mathcal{M}|}\sum_{\mathbf{m}\in\mathcal{M}} d(\mathbf{m}, \mathcal{S}), \qquad d_{\mathrm{H}} = \max\Big\{\max_{\mathbf{s}\in\mathcal{S}} d(\mathbf{s}, \mathcal{M}),\ \max_{\mathbf{m}\in\mathcal{M}} d(\mathbf{m}, \mathcal{S})\Big\}. \tag{3.8}$$

[TODO: Confirm the Chamfer convention used by evaluate_scans (sum or mean of the two directions, squared or unsquared distances) and edit Equation 3.8 to match.] The second Chamfer term penalises missing regions. The Hausdorff distance is set by the single worst point and is very sensitive to outliers, which is why percentiles of $|d_i|$ are also reported.

3.5 Roughness Calculation

Rough Times Ahead

A measured surface combines form, waviness and roughness at progressively shorter wavelengths. ISO 25178-2 defines areal texture parameters on a scale-limited surface, obtained by removing the form and restricting the wavelength band with filters [2, 28]. Here the form is removed by subtracting a fitted low-order polynomial, and the band is set implicitly, at the short end by the grid spacing and MLS radius, and at the long end by the polynomial degree and patch size (section 6.3). A higher degree removes more long-wavelength relief, so the degree decides what counts as roughness. [6] found a planar fit the most robust for sediment in the intertidal region.[TODO: Fact check this]

The parameters, defined in Section section 6.3, fall into three groups. Height parameters (S_a, S_q, S_{sk}, S_{ku}) describe the distribution of residual heights but not their arrangement, so two surfaces can share the same S_q yet have very different feature sizes. Spatial parameters (S_{al}, S_{tr}) describe feature size and directionality. Hybrid parameters combine height and spacing through the slope: S_{dq} is the root-mean-square gradient and S_{dr} the developed interfacial area ratio, the percentage by which the true surface area exceeds the projected area. Rugosity, as used in benthic ecology, is the same quantity expressed as the ratio $A_{\mathrm{true}}/A_{\mathrm{proj}} = 1 + S_{dr}/100$. For gentle slopes the area element $\sqrt{1 + |\nabla z|^2} \approx 1 + \frac{1}{2}|\nabla z|^2$, so $S_{dr} \approx 50\, S_{dq}^2$ % and rugosity is essentially a measure of the mean squared slope.

Scale dependence. Slopes are estimated from height differences between neighbouring grid cells a distance g apart. Natural surfaces have relief over a wide range of scales, often approximately self-affine with a power-law spectrum, so S_{dq}, S_{dr} and rugosity depend on g [23]. Noise makes this worse since uncorrelated height noise σ_n adds a variance of $2\sigma_n^2/g^2$ to a forward-difference slope along each axis. For example, $\sigma_n = 0.5\,\mathrm{mm}$ on a 2 mm grid adds a root-mean-square slope of about 0.35 per axis, which can exceed the true slope of a gently rippled bed. Slope-based parameters are therefore comparable only between surfaces processed with the same grid spacing and smoothing, and the noise contribution must be removed before rugosity is attributed to the sediment. Height parameters are much less sensitive to g.

Chapter 4

System Design

This chapter defines the general requirements of the system as a whole and describes the hardware rig used to capture scans, and the software pipeline built to capture, and process the data to provide useful metrics about the sediment surface. The theory in Chapter [TODO: theory chapter ref] is applied here, and the sensor characterisation in Chapter [TODO: characterisation chapter ref] uses the system as specified below.

4.1 Requirements Definition

The aim of the overarching project is to quantify sandprawn bioturbation on intertidal sediment, within the Langebaan Lagoon, by measuring surface microtopography and extracting roughness metrics and visible biogenic features (mounds and burrow openings). The initial stages of the project were adapted from the framework of Azhar *et al.* [6], primarily modified to use the Intel RealSense D456.

This project aims to provide an investigation into the performance of the RealSense D456 camera currently used. The central aim is to characterise the depth camera in order to determine the optimal hardware and software configuration, and then to use scans taken under those configurations to build a high-quality reconstruction of the sediment on which the analysis is performed. The key output is an analysis of depth noise and reconstruction accuracy under different conditions.

Requirements are grouped as functional (F), experimental (E) and constraints (C), as shown in Table 4.1. Functional requirements state what the system as a whole must do. Experimental requirements state what system variables are investigated in the characterisation. Constraints are fixed conditions imposed on the design.

4.2 System Overview

The system has three main operational stages, shown in Figure 4.1, with a fourth evaluation stage used for characterisation.

1. *Capture*: The capture software rs_marker scans the surface at multiple rotations and camera tilts.
2. *Fusion*: multi-angle scans are cropped to the tag-defined region, aligned and merged into a single reconstruction.
3. *Microtopography processing*: the fused surface is detrended, and roughness metrics and biogenic features are extracted.

For the purpose of characterisation, a fourth evaluation stage is used to gauge the reconstruction and metrics performance of the system against known target relief's.

Table 4.1: System requirements and where they are addressed.

ID	Requirement	Addressed in
F1	Capture short-range depth scans of predominantly planar sediment	Sec. **??**
F2	Restrict analysis to a repeatable, user-defined square region of the surface	Sec. **??**, **??**
F3	Capture the same surface from multiple viewpoints and fuse the scans into one reconstruction	Sec. **??**
F4	Produce detrended microtopography, roughness metrics and segmented mounds and burrows	Sec. **??**
E1	Allow working distance, viewing angle, lighting and depth-filter settings to be varied in a controlled, repeatable way	Sec. **??**, **??**
E2	Support characterisation of the camera's depth noise and error as a function of these conditions, to identify the optimal operating conditions for sediment	Chapter [TODO: ref]
E3	Measure the geometric error of fused scans under these conditions, and identify the optimal configuration for reconstruction	Sec. 4.5
C1	Use a single Intel RealSense D456	Sec. **??**
C2	Rig is portable and adjustable in height for field deployment	Sec. **??**
C3	Results are reproducible and comparable across runs and algorithms	Sec. 4.5



Figure 4.1: System overview. [TODO: Replace with a fuller diagram showing the data formats (.bag, .ply) passed between stages.]

4.3 Hardware Rig

4.3.1 Frame

The existing rig was built from 3030 aluminium extrusion in a 1 m × 1 m × 1 m frame. The top of the frame has a plastic tarp attatched, this allows for the shading of the scan area in the event of direct sunlight degrading the active stereo functionality of the D456. This frame was modified for the purpose of this project, in order to feature an H-shaped cross member that can slide vertically along the four legs. This enables the ability to adjust the camera height, and thus working distance, for purpose of characterisation and easy configuration. It was also modified to feature 3D printed April tag mounts that allow the position to be adjusted

4.3.2 Camera, Tilt and Rotation Mechanism

The Intel RealSense D456 is mounted on the H-frame through a 3D-printed bracket bolted to the extrusion. A servo motor in the bracket tilts the camera by up to $\pm 60^\circ$. The camera clips onto a rotating platform on the tilt stage and can be turned manually through a full 360° in 45° increments. Together, tilt and rotation allow each surface point to be observed from several viewing directions, which reduces occlusion within burrows and mounds. This multi-view scanning additionally averages out view-dependent depth error. The servo is driven by an Arduino microcontroller, connected to and commanded from the operator's laptop, which also runs the capture software. The camera is operated at 1280 × 720 and 30 fps at a nominal working distance of [TODO: optimal distance].

4.3.3 AprilTag Boundary Markers

Four AprilTags are mounted on brackets on the frame legs, and each bracket can slide inward or outward. The tags' positions define a square region on the sediment surface to be processed and analysed. This gives

Table 4.2: Hardware summary.

Component	Specification
Depth camera	Intel RealSense D456
Depth stream	1280×720, 30 fps
Working distance	≈700 mm
Frame	Aluminium extrusion, ≈1 m^3, sliding H-frame
Tilt actuation	Servo, ±60°, Arduino-controlled
Rotation	Manual, 45° increments, 360°
Boundary markers	Four adjustable AprilTags
Ambient light control	Overhead tarp

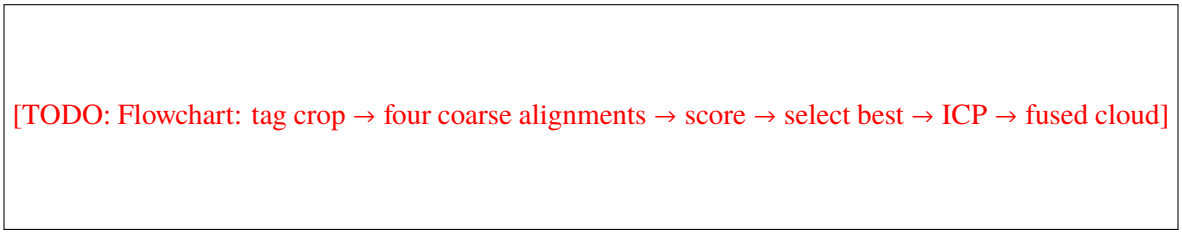


Figure 4.2: Multi-angle fusion pipeline.

a repeatable, physically marked analysis region, and its size can be changed to suit the desired scan region or height configuration of the camera. Because the tags are fixed to the frame, their detected positions also provide a geometric cue for aligning scans taken from different tilt and rotation settings.

4.4 Software Architecture and Processing Pipeline

4.4.1 Capture Software

Capture is implemented in C++ utilising the RealSense SDK. For each rig pose, it applies a configurable chain of depth post-processing filters and records the stream to a `.db3` file, together with IMU data. It detects the four AprilTags in the colour image, and the best frame of each pose is exported as a point cloud in PLY format. [TODO: State the filters used and their parameters, or reference the characterisation chapter.] Tag detection is stored along side captures for use of later processing through cropping and alignment.

4.4.2 Cropping and Multi-Angle Fusion

Each scan is first cropped to the square defined by the four tags. The cropped scans, taken at different tilt and rotation settings, are then fused (Figure 4.2). Fusion needs a good initial pose before refinement, so the pipeline generates several *coarse* alignment hypotheses by independent methods:

1. from the detected tag locations;
2. from the known rotation and servo tilt angle set on the rig;
3. from the camera IMU orientation;
4. from RANSAC matching of FPFH features [**?** **?**].

Each hypothesis is scored with a common alignment-quality measure, and the highest-scoring one is selected. It is then refined with ICP [**?**] and the merged surface passes to the microtopography stage. Using four independent seeds makes the fusion robust to the failure mode of any single method (for example, feature matching on a nearly planar surface, or IMU drift).

4.4.3 Microtopography Processing

The fused surface is detrended to remove the large-scale slope of the sediment, giving the microtopography from which roughness metrics are computed. Mounds and burrow openings are then detected and segmented using classical morphological image processing. [TODO: One sentence each on the detrending method, the roughness metrics, and the segmentation approach; details belong in later chapters.]

4.5 Evaluation Framework and Ground Truth

For characterisation and validation, a separate Python tool, `evaluate_scans`, measures the geometric error of any scan, whether a single capture or the fused output, against a known reference. It reuses the structure of the fusion pipeline's alignment stage, but aligns each scan to the reference. The tool then evaluates scan accuracy and coverage against the reference object.

4.5.1 Reference Surfaces

The reference objects used for validation include a 3D printed egg carton surface $z = sin(x) + sin(y)$, at large and small scale, and a square based pyramid. These reference objects provide a balance of condition matching of the actual sediment surface, using the soft sinusoidal surface, and general performance to less natural sharp shapes. For the purpose of validation, the 3D model files (.stl) are used for evaluation as the 3D mesh surface allows for point to surface distance calculation across the entire 3D object.

4.5.2 Alignment and Metrics

Alignment uses several seeded feature-matching starts, systematic perturbations of the best pose (quarter turns, a flip, and slides of a quarter and half of the part), and point-to-plane ICP, followed by a final refinement against the exact surface. Candidate poses are ranked by a truncated least-squares cost computed in both directions (scan to reference and reference to scan), which prevents a pose that matches fewer points from winning by default. This matters on periodic targets such as the egg carton, where several look-alike poses exist. The tool flags a result as *ambiguous* when the two best poses are too close to call, and reports degrees of freedom the surface cannot constrain.

For each aligned scan the tool reports accuracy (RMSE, MAE, bias, standard deviation, trimmed statistics, percentiles), the vertical error for height-field references, and coverage of the reference. It also reports completeness, precision and F-score at several distance thresholds, Chamfer and Hausdorff distances, outlier fraction and point density. Every run appends a row to a shared table with the exact parameters, reference hash and random seed, so experiments are reproducible and comparable across configurations.

4.6 Summary

The rig provides repeatable, multi-angle capture of a user-defined surface region. The software fuses those scans, extracts microtopography, and is checked against analytic ground truth. Chapter [TODO: characterisation chapter ref] uses this system to characterise the D456's depth noise.

Chapter 5

Intel RealSense Characterisation

The aim of the overarching project is to produce high-accuracy reconstructions of intertidal sediment from which surface roughness metrics can be extracted. The accuracy of those metrics is limited by the depth data they are computed from, so the Intel RealSense D456 was characterised before the capture and processing pipeline was finalised. The characterisation answers four practical questions about how the camera and rig should be operated in situ: at what working distance, at what viewing angle, under what ambient lighting, and with which depth-stream and filter settings the camera produces the most accurate surface.

The characterisation was performed in two stages. In the first stage the camera was tested against flat planar targets. A plane has an exact, easily measured ground truth, which allows the effect of working distance, tilt angle and ambient light to be isolated from the effect of surface shape. These tests give a baseline for sensor noise and provide the operating envelope used in the second stage. In the second stage the camera was tested against 3D-printed reference objects with known geometry (see Appendix C.) These tests identify the camera configuration, post-processing filter settings and combination of viewing directions that give the best reconstruction, using surface relief that is representative of the sediment of interest. Each characterisation below is presented with its method, results and a short discussion.

5.1 Plane Surface Characterisation

As Plain As That

5.1.1 Capture procedure and metrics

All planar characterisation used custom C++ software built on the Intel RealSense SDK, following the plane-fit metrics of Intel's Depth Quality Tool. Details of each trial, such as distance, tilt angle and a lux reading, are entered into the tool and the scan is started. A region of interest (ROI) is then applied whose pixel size is adjusted with distance so that it always covers a fixed 50×50 cm area of the target. A plane is fitted to the ROI points of every frame and compared with the ground-truth plane. Several metrics are reported by the tool, however there are four main metrics that will be discussed here to represent the general performance, each averaged over the valid frames of a trial.

- **Median depth error** (mm): the median signed difference between the measured and ground-truth depth, which measures systematic bias.
- **Depth RMSE** (mm): the root mean square representing depth noise from camera to plane.
- **Plane-fit RMS** (mm): the root-mean-square distance of the ROI points from the best-fit plane. This measures spatial noise and is independent of any constant offset in the depth.
- **Valid Pixels** (%): the proportion of ROI pixels that returned a valid depth value.

The distinction between precision and bias matters for this project. A smooth, systematic bias is removed when the fused surface is detrended to extract microtopography, whereas random noise adds directly to the measured roughness. Plane-fit RMS and fill rate are therefore treated as the primary metric, with depth error and depth RMSE treated as secondary indicators. Each condition was recorded as a single trial. The frames within a trial capture temporal variation but not the repeatability of re-setting the experiment, which is discussed in **??**.

5.1.2 Planar Characterisation Methodology

Working distance

The camera was mounted on a tripod and pointed at a white, featureless wall, as recommended by Intel for depth-quality testing. This forces the camera to depend on its own projected infrared pattern. The tripod was used instead of the experimental rig because the rig's mounting brackets could not hold the camera sufficiently parallel to the floor: the camera angle varied with every scan and consistently exceeded 1.5°, which would have contaminated a distance sweep. The camera was positioned at eight distances between 604 mm and 903 mm, and the angle between the camera and the wall was minimised by hand at each position. The fitted-plane angle confirmed that the residual pitch and roll stayed below 0.5°in all trials. The ground-truth distance was measured between the wall and the cameras glass front using a tape measure.

[TODO: Maybe talk more about methodology?]

Camera tilt

The camera was mounted on the experimental rig so that its angle could be set with the tilt servo. A large flat white foam board was placed on the floor and re-centred in the field of view (FOV) after each change of angle. Two rotations were tested independently. The pitch axis is parallel to the line through the two IR imagers, and the roll axis is perpendicular to both the pitch axis and the optical axis. The angle was changed in steps of approximately 2°, covering pitch angles from $-27°$ to $-2°$ and roll angles from $+3°$ to $+25°$ for 26 trials in total. The camera mount height was kept fixed. For each trial the vertical height of the left IR imager, which is the centre of the resulting depth image, above the board was measured, and the axial ground-truth distance was calculated from that height and the tilt angle. This axial distance varied between 657 mm and 743 mm over the sweep. The angle of the fitted plane agreed with the set angle to within 1° in every trial (mean 0.4°), which verifies the angle settings.

[TODO: Maybe talk more about methodology?]

Ambient lighting

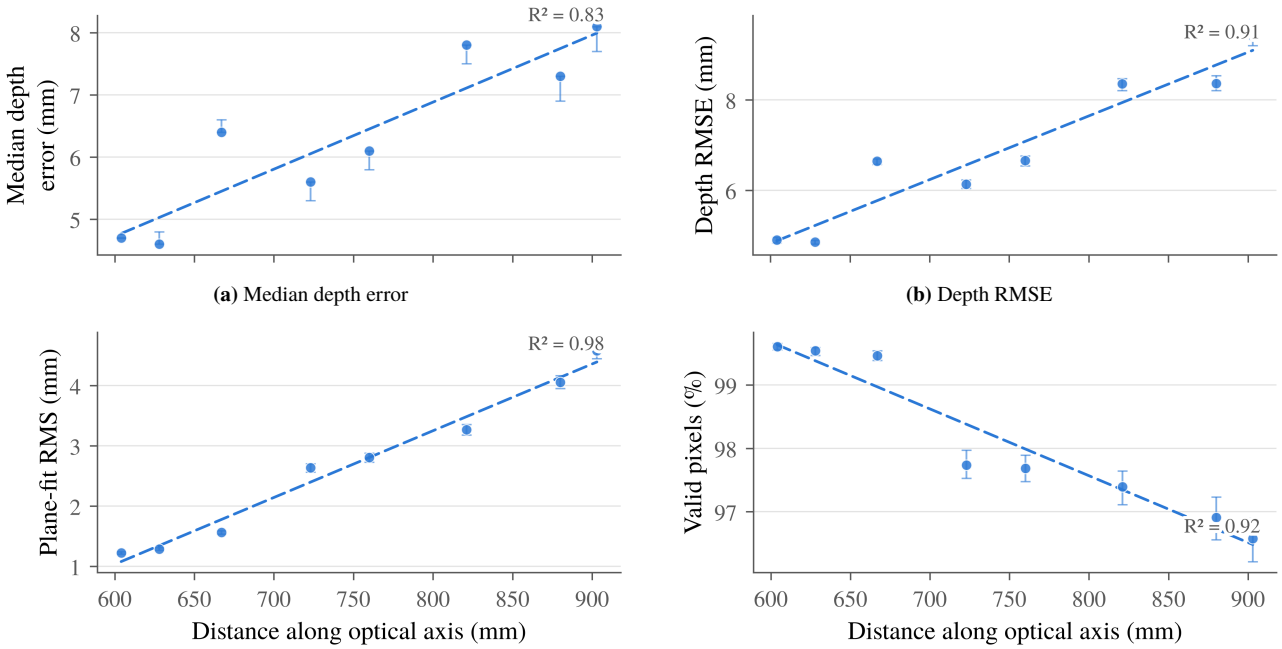
The same foam board was scanned at a camera height of approximately 900 mm under a range of lighting conditions, with the camera exposure left on automatic. Indoor conditions ranged from a dark room in the evening (blinds closed, lights off) to an afternoon room with the blinds open and the lights on. Outdoor trials were performed around midday in shade, in shade within direct sunlight, and in direct sunlight. The outdoor trials were taken twice per condition, one with the foam board, and one without the foam board. This was done as the effects of the high ambient light resulted in exposure settings that were unable to detect the cameras built in projected pattern, and thus the featureless white surface lost detection. The brick flooring was able to be detected in this condition, thus the results for the outdoor trials provide relational performance compare to the rest of the trials. Illuminance was read with a smartphone running the Physics Toolbox Suite application, which uses the front camera as a lux meter. Such readings are uncalibrated and are used here to indicate the order of magnitude and ranking of conditions, not precise values. The lux was recorded as 0 in the darkest condition and 24 400 in full sun.

[TODO: Maybe talk more about methodology?]

5.1.3 Planar testing Results

Working Distance

Results for planar distance characterisation are shown in Table D.1. These are then plotted in Figure 5.1.



(a) Median depth error

(b) Depth RMSE

(c) Plane-fit RMS

(d) Valid pixels

Figure 5.1: Plots Showing Planar Performance Against Height. Per-trial Median with IQR Over n = 385–552 Frames.

Camera Tilt

Results for planar angle characterisation are shown in Table D.2. These are then plotted in Figure 5.2.

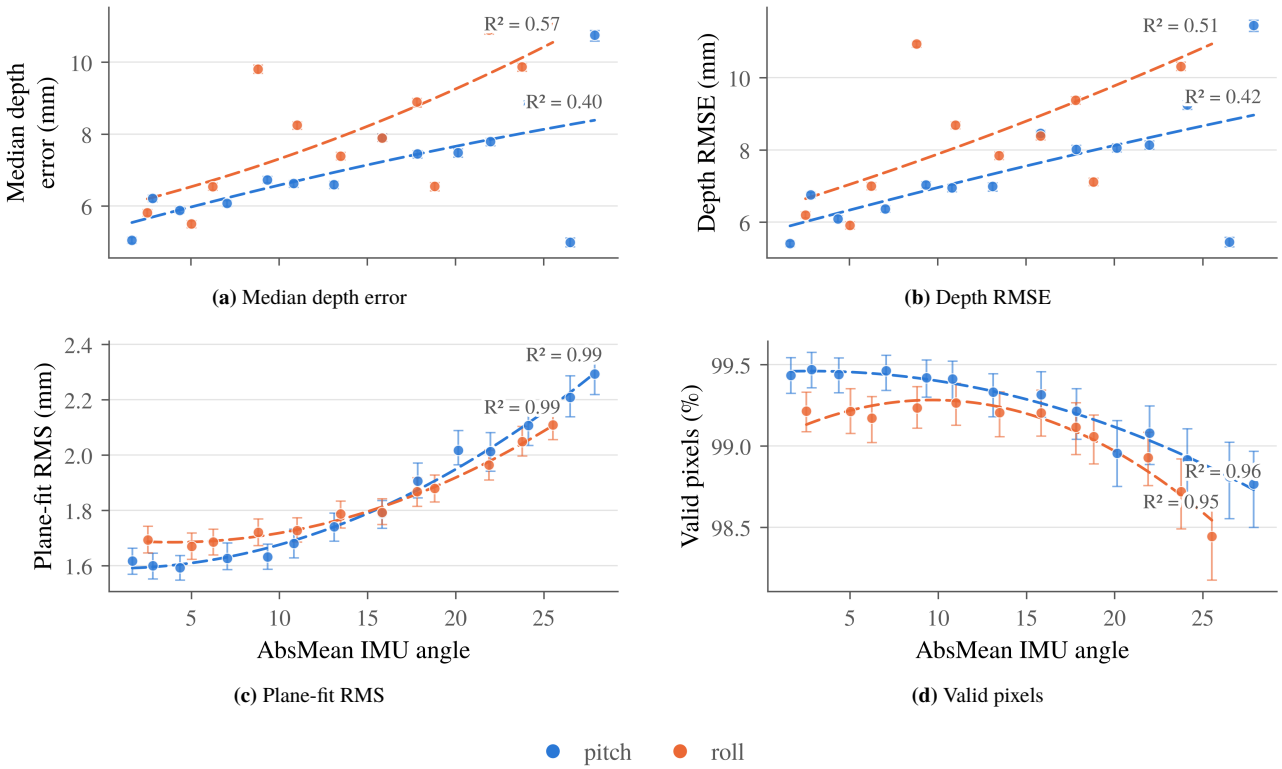
Ambient Lighting

Results for planar angle characterisation are shown in Table D.3. These are then plotted in Figure 5.3. Lighting conditions with corresponding illumination values are shown in Table 5.1

5.1.4 Discusion

Working Distance

Stereo noise is expected to grow with the square of the range, however as can be seen in Figure 5.1, the data is closer to a linear trend. Part of the difference may come from the ambient light, which was not completely constant across the sweep, as trials were taken across a time period between 15:40 and 17:35, where the auto-exposure value dropped from around 5300 for the three closest distances to between 1700 and 2050 for the



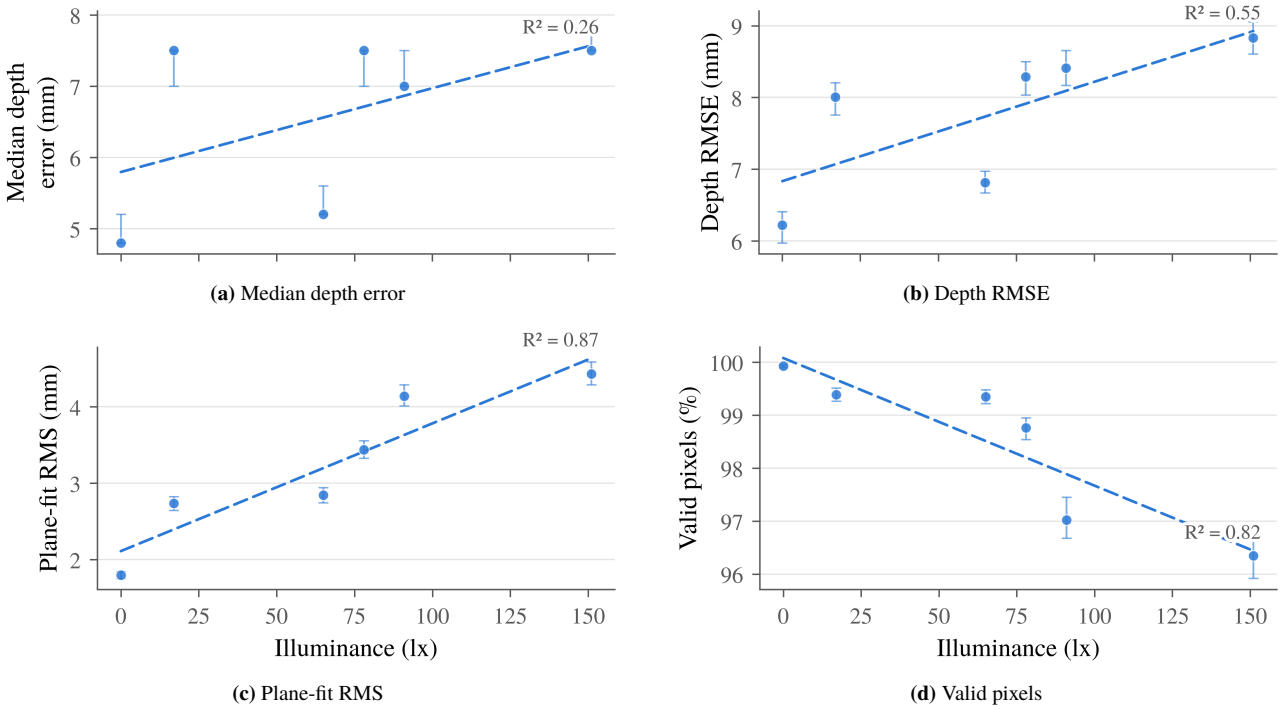
(a) Median depth error

(b) Depth RMSE

(c) Plane-fit RMS

(d) Valid pixels

Figure 5.2: Plots Showing Planar Performance Against Scan Angle. Per-trial Median with IQR Over n = 527–574 Frames.



(a) Median depth error

(b) Depth RMSE

(c) Plane-fit RMS

(d) Valid pixels

Figure 5.3: Plots Showing Planar Performance Against Light Intensity, for Indoor Conditions with Foam Board. Per-trial Median with IQR Over n = 592–599 Frames.

Table 5.1: Planar characterisation under different lighting conditions at a height of approximately 0.9 m. Lux values are indicative (smartphone sensor).

Surface	Condition	Lux
Indoor foam board	Evening, blinds closed, Florescent lights off	0
	Afternoon, blinds closed, Florescent lights off	17
	Evening, blinds closed, Florescent lights on	65
	Afternoon, blinds closed, Florescent lights on	78
	Afternoon, blinds open, Florescent lights off	91
	Afternoon, blinds open, Florescent lights on	151
Outdoor foam board	Shade	388
	Direct sun, under shade	2 650
Outdoor, no board	Shade	213
	Direct sun	9 800
	Direct sun, under shade	10 000

remainder. However another reason for the linear appearance is likely due to the small distance variation, revealing a locally linear like region, compared to the larger exponential shape. Even allowing for this, there is a useful trend that can used for this project. Depth noise shown i Figure 5.1a grows with working distance, simultaneously while the percentage of valid pixels shown in Figure 5.1d declines. This shows that the camera should generally be operated as close to the target as feasible to minimise absolute errors. By looking at the depth error as a percentage of actual height, we see that there is a slight upward trend from 0.77% to 0.90%, however there is significant variance in the trend, with an trend fit score of only 0.18, thus the approximately constant relative depth error is consistent with a bias or scale, rather than a distance dependant effect.

Camera Tilt

Stereo noise is expected to follow a sinusoidal relationship between the noise and the tilt angle. Whilst not completely sinusoidal, it can be seen in Figure 5.2c and Figure 5.2d, that there is a similar relationship present with the plane fit RMS and percentage of valid pixels. The trends fitted to these data are however quadratic, as to avoid over fitting with a higher order polynomial. Whilst there is a weak trend in Figure 5.2a and Figure 5.2b, Figure 5.2c and Figure 5.2d both show that scans taken at lower tilt angles, generally are able to higher quality planar scans, compared to higher angles of incidence. One consideration that must be taken here, is that axial distance varies with tilt angle, and as such the results seen here are compounded with performance of distance, and as such these do not represent absolute angle performance. However this is accepted for the purposes of this study, where the cameras operating height remains static when scanning sediment, thus this result better reflects the performance of the camera in this specific use case.

Ambient Lighting

Two conclusions follow. First, ambient light degrades the depth from active stereo even when the surface is well within the sensor's range, because the projected infrared pattern has to compete with the ambient light for contrast. Daylight appears more damaging per lux than artificial light which can be seen in Figure 5.3c and Figure 5.3d, with 4.14 mm RMS and 97.0% valid pixels at 91 lux of daylight against 2.84 mm RMS and 99.3% valid pixels at 65 lux of artificial light, which is consistent with the greater near-infrared content of sunlight, so illuminance in lux is only a rough predictor of performance. Second, the surface matters as much as the illumination. The board completely failed at 2 650 lux in outdoor sun exposed areas, whereas the ground surface returned complete, low-noise data at 9 800 lux, so the failure cannot be attributed to the light level alone. A bright, featureless, highly reflective white surface is a worst case for stereo matching. The ground surface supplies natural texture, so the planar results on white targets should be read as pessimistic bounds on the noise expected on sediment. The outdoor tests did not include wet or glossy sediment, which is more specular, and

thus could influence results more. These results support retaining the shading tarp on the rig, to limit the ambient infrared reaching the scan area.

5.2 3D Reference Characterisation

A Whole New Dimension

The planar tests establish the sensor's noise behaviour but not how accurately it reconstructs three-dimensional relief, which is affected by resolution, filtering, distance and the fusion of several views. These effects were measured against reference objects with known geometry using a modified version of the main capture program, which records scans under controlled software settings and hardware set-ups.

5.2.1 Reference objects and evaluation

Three 3D-printed reference objects were used (chapter C). Each has a 100×100 mm footprint. Two are sinusoidal "egg carton" surfaces of the form $z = \sin(x) + \sin(y)$ at a large scale (feature spacing 50 mm, amplitude 20 mm) and a small scale (25 mm, amplitude 15 mm), and the third is a square-based pyramid of 50 mm height. The smooth sinusoidal surfaces are analogous to the smooth mounds produced by sand prawns, whose heights of 3 to 5 cm were reported in the literature review, and the small carton tests the lateral resolution of the camera. The pyramid has sharp edges and planar facets, which show whether the filters preserve geometric edges or smooth them.

Scans were evaluated with the `evaluate_scans` tool described in section 4.5. The tool rigidly aligns each scan to the reference .stl model, using feature-matching starts followed by point-to-plane ICP, and then computes point-to-surface distances. Scale is not estimated, so any scale error in the camera data appears as geometric error. The metrics used are Accuracy RMSE, signed bias, completeness at 2mm and computed F-score at 2mm. F-score is a combined metric which aims to represent both precision and completeness.

$$P(\tau) = \frac{1}{|S|} \sum_{s \in S} \mathbb{1}\Big[\min_{r \in R} \|s - r\| < \tau \Big] \tag{5.1}$$

$$C(\tau) = \frac{1}{|R|} \sum_{r \in R} \mathbb{1}\Big[\min_{s \in S} \|r - s\| < \tau \Big] \tag{5.2}$$

$$F(\tau) = \frac{2\,P(\tau)\,C(\tau)}{P(\tau) + C(\tau)} \tag{5.3}$$

Let S denote the set of scanned points and R the set of points sampled from the reference model. For a distance threshold τ, precision Equation 5.1) is the fraction of scanned points lying within τ of the reference surface, and completeness (Equation 5.2) is the fraction of reference points lying within τ of the scan. The F-score (Equation 5.3) is their harmonic mean, and therefore penalises both spurious points and missing coverage. It is calculated at $\tau = 0.5$, 1, 2 and 5 mm, however for the purposes of this report, all results consider the threshold of $\tau = 2$ mm as this gives a balance of high accuracy representation, whilst still including relatively large bias/point spread.

5.2.2 Testing Procedure

Camera configuration

The first sweep fixed the height and tilt of the camera and scanned the large egg carton and the pyramid with different stream configurations. The resolution ranged from 1280×720 to 640×360 pixels, the depth units from 1 mm to 0.1 mm, and all of the built-in visual presets were tested except the Hand preset, which is intended

for gesture tracking. The depth unit sets the quantisation step of the 16-bit depth values, and the finest setting (0.1 mm) still provides a maximum range of 6.5 m, well beyond the working distance.

Based on this sweep, a resolution of 1280×720, a depth unit of 0.1 mm and the High Density preset were used for all later scans[TODO: confirm that this was the outcome of the sweep]. These are the settings listed in chapter 4.

Post-processing filters

With the configuration fixed, five stationary scans of each object were recorded at each of the tested heights, without tilt or rotation. The data were recorded to files so that the RealSense post-processing filters could be applied offline in many combinations to the same scans. The filters investigated were decimation, spatial, temporal and hole-filling, and a sweep was made over their smoothing alpha, smoothing delta and magnitude parameters[TODO: list the values or ranges tested for each filter]. Decimation reduces the resolution, the spatial filter performs edge-preserving smoothing, the temporal filter averages depth across frames, and hole filling interpolates invalid pixels. Each combination was evaluated against the reference, and the process was repeated at each height to determine whether the best filter settings depend on the working distance.

Tilt and rotation

Finally, a full scan of each object was recorded at rotations of $-45°$, $0°$, $45°$ and $90°$ and at tilts from $-25°$ to $+25°$ in steps of $5°$. This gives 44 views of each object. These views were aligned and fused by the pipeline in chapter 6. Subsets of the views were fused and each fusion was evaluated against the reference, which shows how the reconstruction accuracy depends on the number of views and on the combination of tilt and rotation.

5.2.3 3D Characterisation Results

Camera Configuration

Results for 3D parameter characterisation are shown in Table D.4, Table D.5 and Table D.6. These are then plotted in Figure 5.4, Figure 5.5 and Figure 5.6 respectfully.

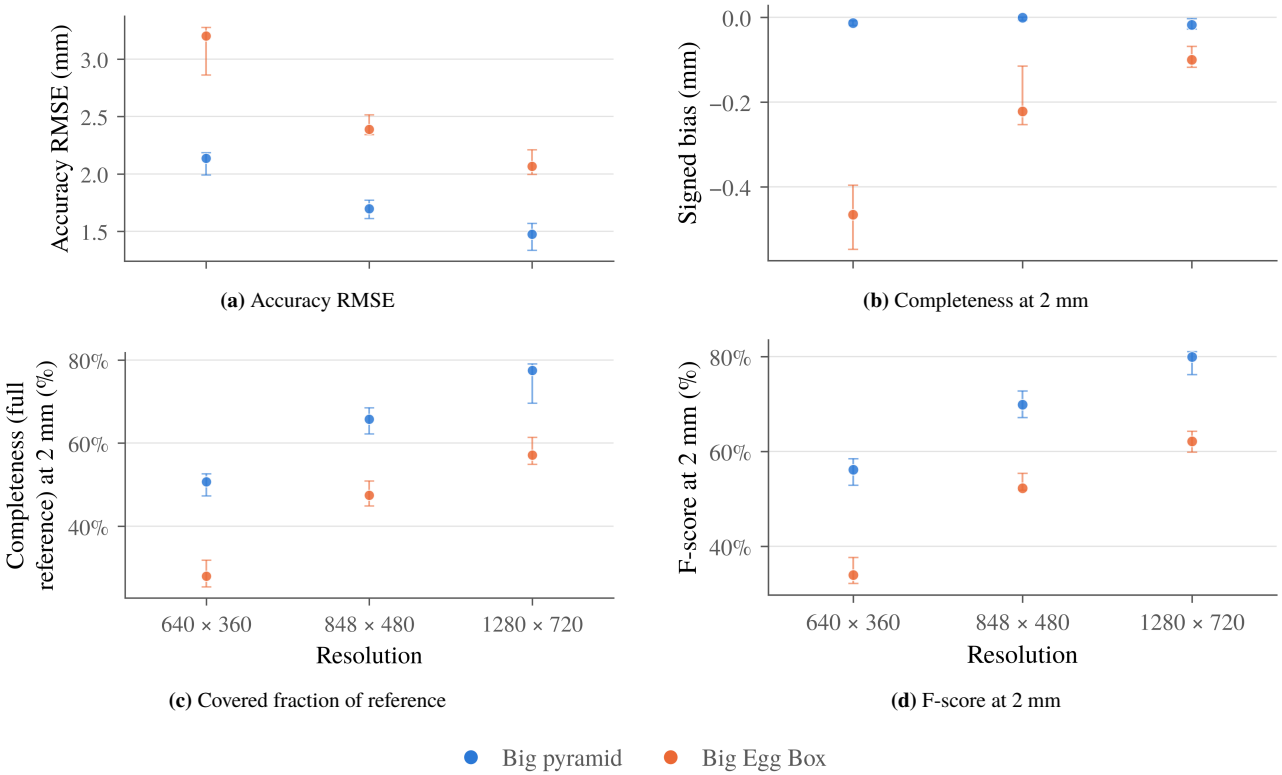
A full heatmap describing the performance of all parameter combinations can be found in section E.3.

Filters

A full comparison of all filter parameters is infeasible for the scope of this report. Thus only a comparison between a baseline evaluation, with no filters applied, and the best performing filters is shown in Figure 5.7, with the data shown in Table D.7. The filter settings can be found in Table 5.2

Table 5.2: Post-processing filter settings of the best-performing configuration for all objects combined and for each object.

	Decimation	Temporal Filter			Hole Filling	
Object	Magnitude	α	δ	Persistence	Mode	Spatial Filter
All objects	Disabled	0.1	20	Disabled	Fill From Left	Disabled
Big Egg Box	2	0.1	20	Disabled	Fill From left	Disabled
Small Egg Box	Disabled	0.4	20	Valid in 2/8	Disabled	Disabled



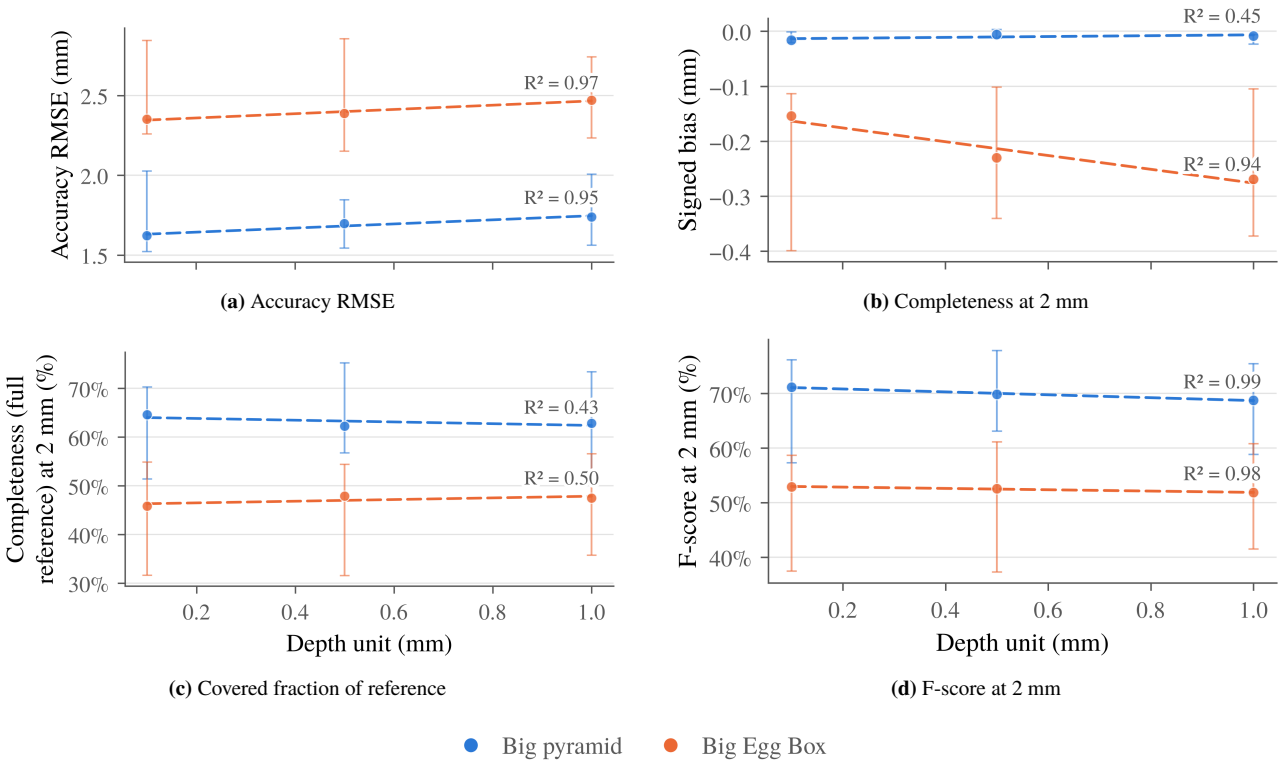
(a) Accuracy RMSE

(b) Completeness at 2 mm

(c) Covered fraction of reference

(d) F-score at 2 mm

Figure 5.4: Plots Showing 3D Performance Against Camera Resolution. Median with IQR of n = 15 scans per point, Taken Over Three Depth Units, and Five Presets.



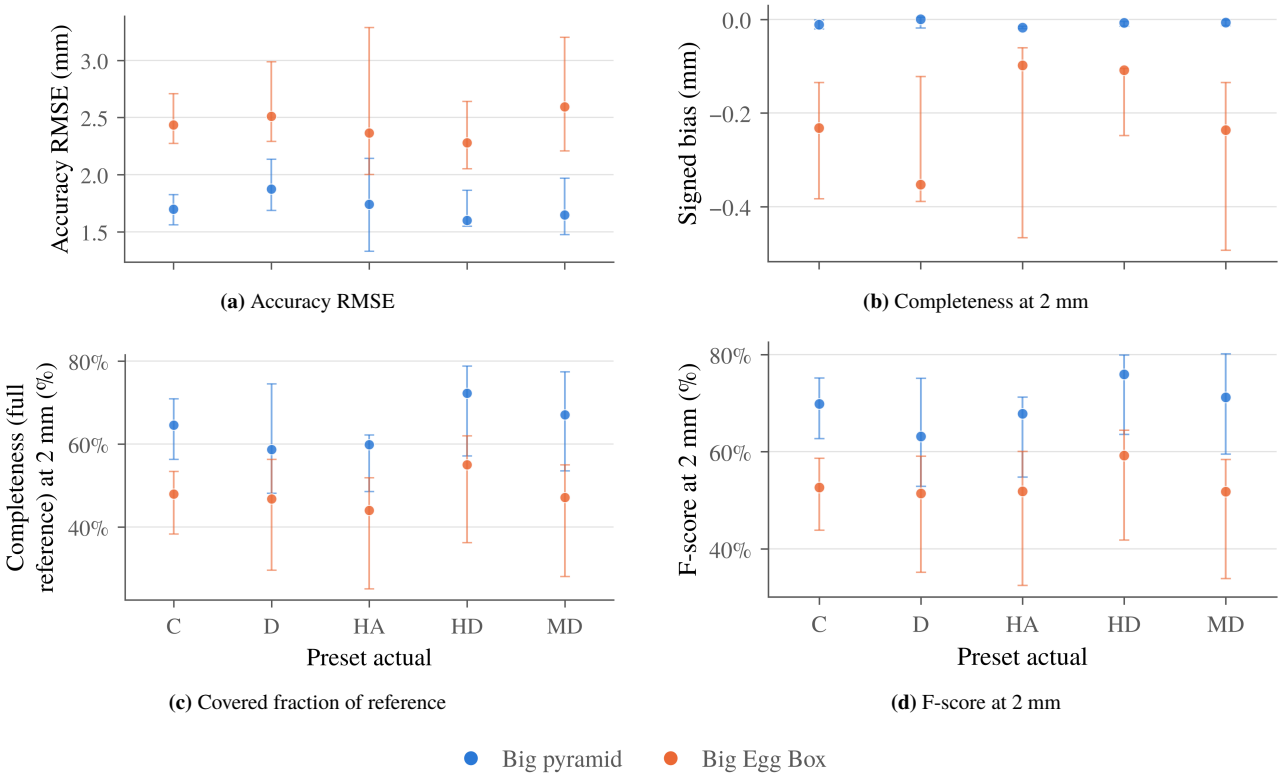
(a) Accuracy RMSE

(b) Completeness at 2 mm

(c) Covered fraction of reference

(d) F-score at 2 mm

Figure 5.5: Plots Showing 3D Performance Against Depth Units. Median with IQR of n = 15 scans per point, Taken Over Three Resolutions, and Five Presets.



(a) Accuracy RMSE

(b) Completeness at 2 mm

(c) Covered fraction of reference

(d) F-score at 2 mm

Figure 5.6: Plots Showing 3D Performance Against Camera Preset. Median with IQR of n = 15 scans per point, Taken Over Three Depth Units, and Three Resolutions.

Tilt and Rotation

Again, a full comparison of all metrics agsainst all variables is infeasible for the scope of this project. The results shown in Table D.8 are plotted in Figure 5.8, and describe the comparison between a fusion of every tilt and rotation, compared to a single scan. This combination was chosen as it represents the best overall performance of the system as a whole.

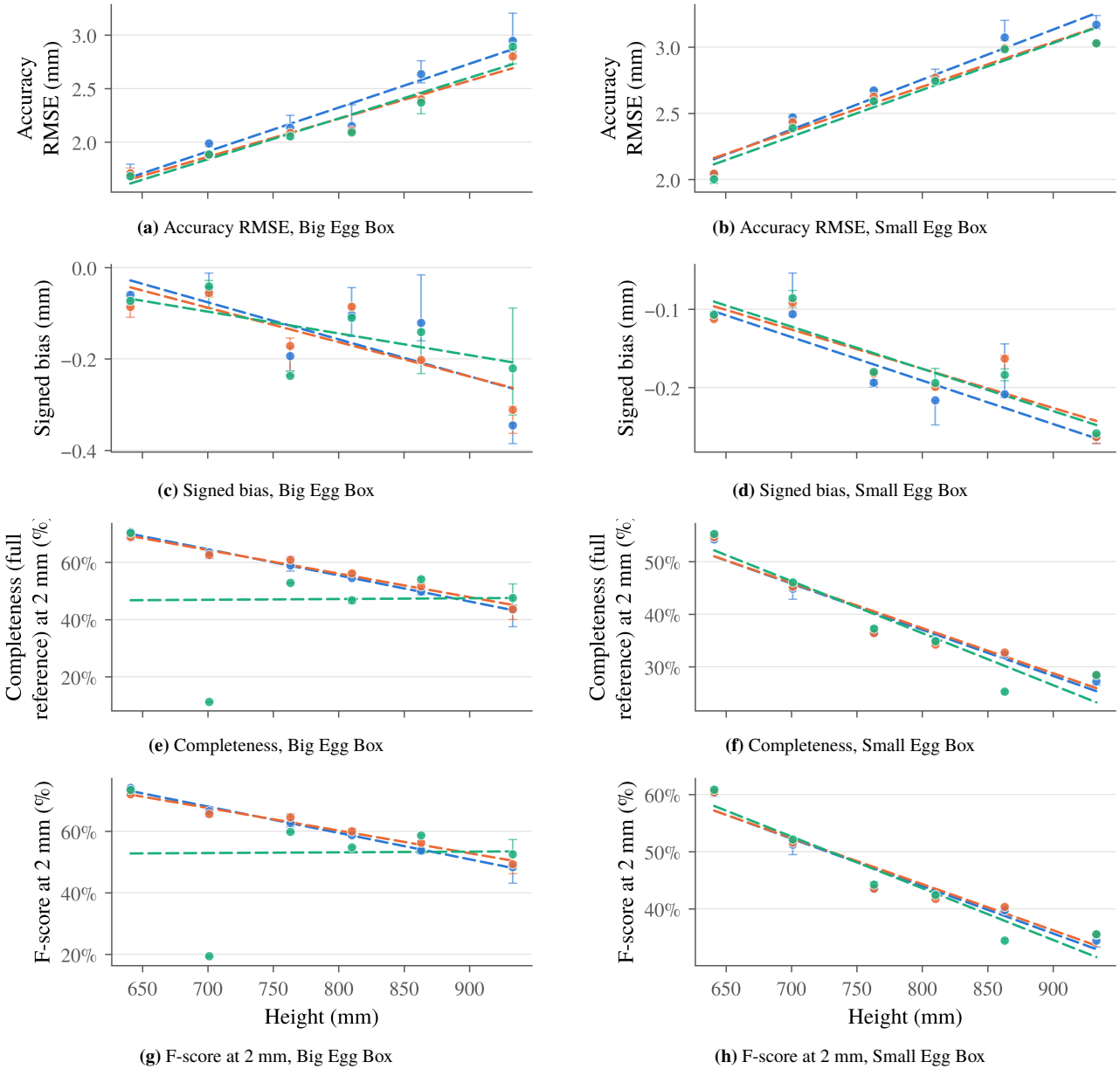
A More in depth comparison of the different combinations can be seen in section E.2.

5.2.4 Discussion

Camera Configuration

Out of all results shown for the different camera calibrations, the comparison between depth image resolution and accuracy shown in Figure 5.4 shows the trend. For the egg box surface, it can be seen that a higher depth resolution results in a higher performance. Going from the default resolution of 848 × 480, to the highest supported resolution of 1280 ×720 results in an RMSE that drops by 0.32mm, or a decrease of 13.4%, and a completeness that increases by 9.6%. An interesting observation is that the results for the large pyramid are generally much better than that of the large wave, which is contrary to expectation, as the generally more geometrically sharp features of the pyramid tend to get smoothed much more than the smooth features of the egg box surfaces. This is likely due to the egg box shape having a higher feature density than the pyramid, resulting in more accuracy loss.

Results from depth unit show that there is a slight trend in the data, tending to higher performance at lower depth units, which makes sense, however the result difference is smaller than expected, especially considering the results for the large pyramid, which was expected to have more significant results changes, due to the improved quantisation of the depth, on angled planar surfaces. Generally the best performance is achieved with the



(a) Accuracy RMSE, Big Egg Box

(b) Accuracy RMSE, Small Egg Box

(c) Signed bias, Big Egg Box

(d) Signed bias, Small Egg Box

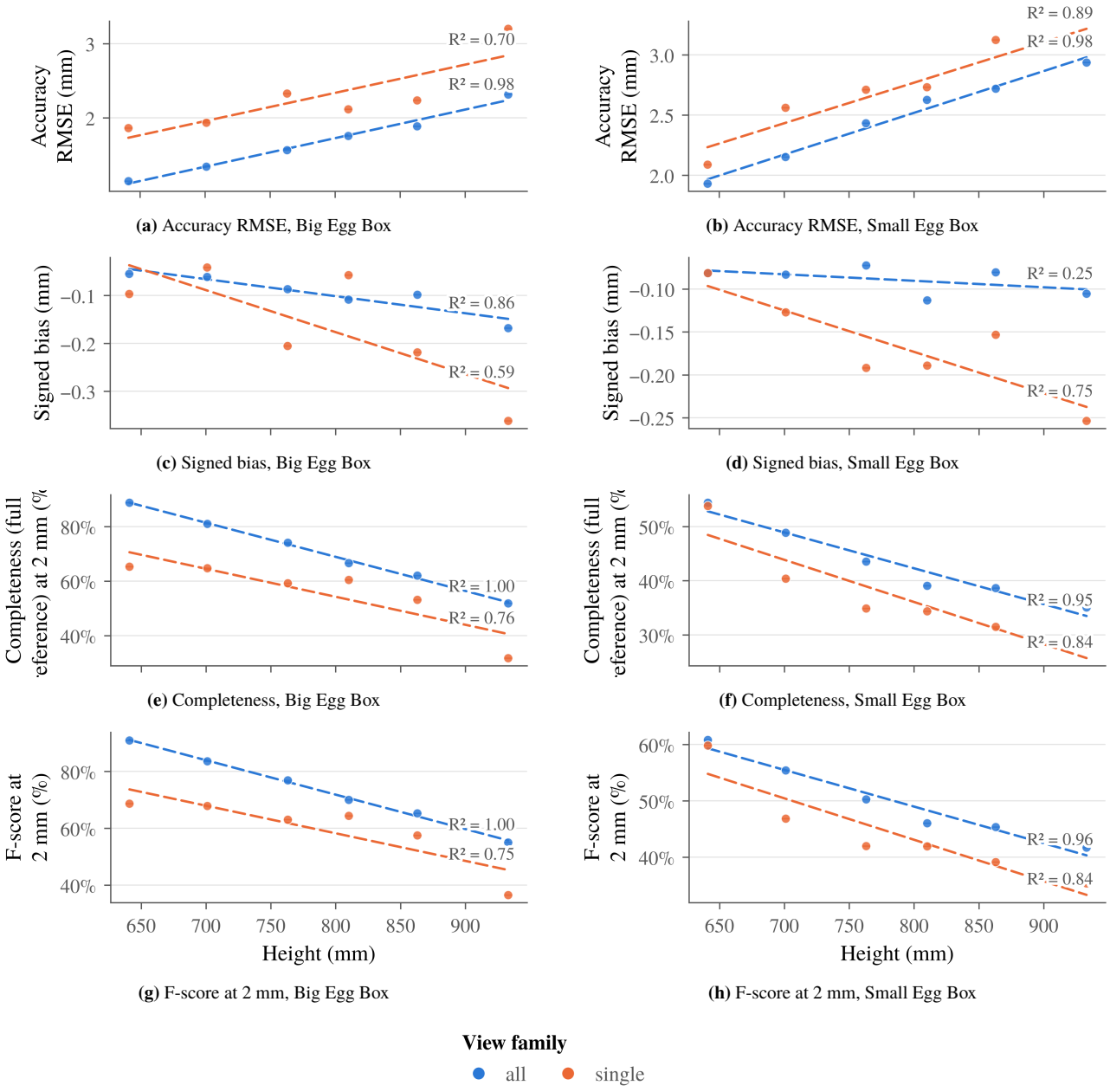
(e) Completeness, Big Egg Box

(f) Completeness, Small Egg Box

(g) F-score at 2 mm, Big Egg Box

(h) F-score at 2 mm, Small Egg Box

Figure 5.7: Plots Showing 3D Performance Comparison Between No Filter and Best Performing Filter, Best Fit (R^2 0.00 to 1.00). Median with IQR of n = 4–5 Scans Per Point.



(a) Accuracy RMSE, Big Egg Box

(b) Accuracy RMSE, Small Egg Box

(c) Signed bias, Big Egg Box

(d) Signed bias, Small Egg Box

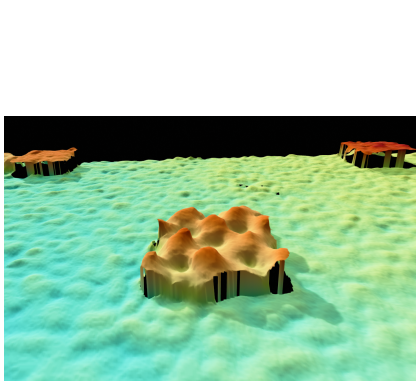
(e) Completeness, Big Egg Box

(f) Completeness, Small Egg Box

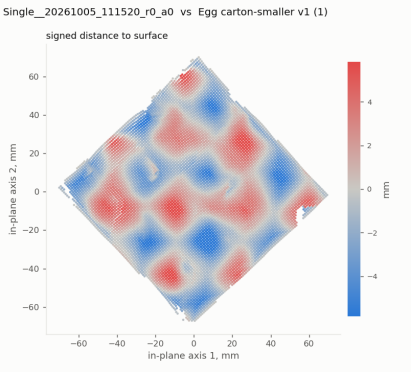
(g) F-score at 2 mm, Big Egg Box

(h) F-score at 2 mm, Small Egg Box

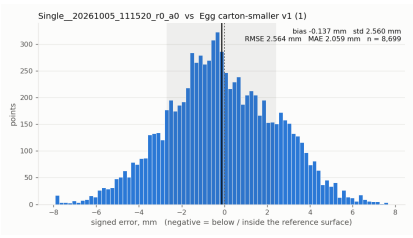
Figure 5.8: Plots Showing 3D Performance Between No Fusion and All View-Point Fusion. Median with IQR of n = 1 Point Clouds Per Point.



(a) 3D Mesh of Single Scan Viewed From Side

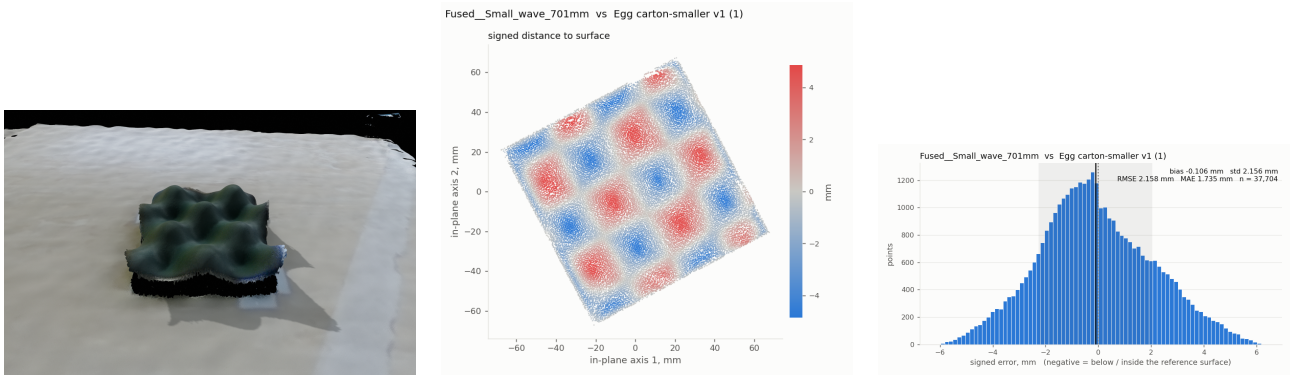


(b) Evaluation Showing Error Map of Single Scan Compared Against Reference Object



(c) Histogram Showing Spread of Signed Error

Figure 5.9: Visual Representation of Single Scan Performance Using Small Egg Box Surface



(a) 3D Mesh of Fused Scan Viewed From Side

(b) Evaluation Showing Error Map of Fused Scan Compared Against Reference Object

(c) Histogram Showing Spread of Signed Error

Figure 5.10: Visual Representation of Fused Scan Performance Using Small Egg Box Surface

minimum depth unit scale of 0.1mm, however larger depth units are likely able to produce very similar results

The highest performing preset was shown to be high density, which achieved the highest metrics across both shapes. This result does make sense, as the high density preset tends to maximise the resulting pointloud, which produces less depth holes in the scan, which help to provide higher accuracy reconstructions. Further consideration needs to be taken to investigate the performance of this setting as the multi-view fusion will likely have a large effect on this. It is expected that high accuracy will likely proove to be the best performing preset, given that the multi-scan fusion will resolve large depth holes in the scan.

Filters

The analysis of different filters show that there is a minimal benefit from applying these filters. The best performing filters are described in Table 5.2. inspecting more detailed plots shown in section E.1, show that there is a general trend finding that more aggressive temporal filtering, less agressive spatial filtering and minimal decimation improve the results the most. However these results are generally dwarfed by the effects of height, as can be seen in these plots, where accuracy variation across a single hight is less than between heights. These results differ from what is expected, since it was expected that a very agressive temporal filter would help to reduce temporal noise which generally results in high variance, especisally in this case where each scan is static. A possible reason for this, could be the fact that each reference object is reletivly small, being effected less by large scale temporal noise, or that temporal noise generally averages out to zero.

It can be seen though, that decimation results in a minimal degredation in the accuracy, potentially leading to processing performance, by decimating the scans to lower resolutions after the fact. However this is still not recommended, as large area scans are likely to suffer from large information loss.

Tilt and rotation

it can be seen in Figure 5.8 that multi-view fusion generally always improves the accuracy of the scans taken. This fusion of all different viewpoints increase the f-score of the resulting fusion by an average of 14.0% for the big egg box surface and by 5.77% on the small egg box surface. This shows that fusion is a viable way to add extra accuracy to a scan without requiring differing hardware. by inspecting Figure 5.9c and Figure 5.10c, it can be seen that whilst the fusion process does not significanly reduce the RMSE or standard deviation of the reconstruction, but it does smooth out the distribution significanylu, and looking at the error maps and 3D meshes, the fused scan produces a much smoother reconstruction, that much closer matches the reference object. This smoothing is specifically a very important metric. Even if the absolutre accuracy is only slightly improved, the roughness metrics of the single scan should be drastically larger than that of the fused scan, showing that a single scan may skew results. This can also be seen by looking at the floor surface surrounding the object. The

single scan features large undulations and variability across the flat surface, whilst the fused scan has a much more consistently flat surface

Chapter 6

Data Processing Pipeline

The capture stage (chapter 4) produces one session per experiment. A session contains multiple individual scans of the same patch of sediment, one per combination of mount rotation and tilt. Each scan stores the single best depth frame as a coloured point cloud (`.ply`), together with the AprilTag detections for the whole recording, the IMU sample taken with the best frame, the full recording and a text dump of the camera configuration.

A single scan of the surface contains several factors that result in lower quality reconstructions. It contains the rig and regions outside of the frames bound due to the wide field of view of the camera. It is noisy, particularly where the surface is seen at a grazing angle. It also has holes wherever relief hides part of the surface from that viewpoint. The processing pipeline turns a session of multiple of such scans into a single, clean reconstruction of the region of interest and then into numbers that describe its roughness. There are four main stage in the data processing pipeline.

1. **Cropping** (section 6.1) cuts every scan down to the square of sediment bounded by the four corner AprilTags.
2. **Fusion** (section 6.2) registers the cropped scans into one common coordinate frame and merges their overlapping regions into a single, lower-noise surface with fewer holes.
3. **Roughness estimation** (section 6.3) levels and detrends the fused surface and computes standardised surface-texture parameters, including rugosity.
4. **Feature detection** (section 6.4) locates and counts discrete features such as sandprawn burrow openings.

Each stage is a separate program that reads the previous stage's output files, so any stage can be rerun or inspected on its own. The language of each program was chosen for its workload. Cropping is mostly bookkeeping and image geometry, so it is written in Python with Open3D and NumPy. Fusion runs nearest-neighbour searches, feature matching and iterative registration on millions of points, so it is written in C++ with the Point Cloud Library. The roughness metrics are array operations on a regular grid and are written in Python with NumPy and SciPy. A small Python script (`bulk_process.py`) chains cropping and fusion over many sessions without user interaction.

6.1 Point Cloud Cropping

In Too Deep

The raw point clouds have three problems that make them unsuitable for direct use.

1. **The field of view is larger than the region of interest.** The Intel RealSense D456 has a wide field of view, so even with the camera pointed straight down it records depth well outside the patch of sediment inside the experimental rig.

2. **The view changes with every scan.** Because the camera is carried on a rotating and tilting mount, tilted scans look past the edge of the rig and often capture the rig frame itself. These points do not belong to the surface and would corrupt both registration and the roughness statistics.

3. **Large scans are expensive and overlap poorly.** Fusion compares every scan with every other scan already placed. Points outside the shared region add computation without adding information.

The cropping step therefore reduces every scan to the same physical region: the square on the sediment bed whose corners are marked by four AprilTags. Because the tags are fixed in the world, this region is the same in every scan regardless of how the camera was rotated or tilted. This gives fusion a consistent, well-overlapping input. The step is implemented in `crop_pointclouds.py`, which processes an entire session at once. It crops automatically wherever it can and falls back to a manual crop when it cannot.

6.1.1 Finding the tags in each scan

AprilTag detection is performed during capture on the colour image, and every detection is written to a `.csv` file together with the frame number, the tag ID and the pixel coordinates of the tag's centre and four corners. Squares that look like tags but whose ID could not be decoded are also logged, with an ID of -1.

The point cloud of a scan comes from one particular frame of the recording. The cropper reads that frame number from the scan's IMU file, which was written at the same moment as the point cloud. It then collects the detections from that frame, however tag detections are accepted from ± 15 frames around the best frame, since each scan is taken from a static position. This allows for some redundancy against frames where detections momentarily failed. Decoded tags are always preferred, for region identification, however undecoded squares are used to fill corners where no identified tags are visible, and only when they lie close to where the missing tag is expected.

6.1.2 Learning the tag layout of a session

Not every tag will be detected, or visible in each scan wi thin a session. Glare, shadow or a steep tilt often hide tags. Therefore, in order to maintain the ability to automatically crop each scan to the correct region, the program first learns the layout of the tags from the scans in the session where all four were detected. From those frames it records:

- The order in which the four tag IDs appear going around the square, which says which tags share a side and which are diagonal.
- The ratio of the square's side length to a tag's side length.
- The aspect ratio of the square, i.e. how each side compares to its neighbours.

The median of each quantity over all four-tag frames is used, which makes the layout robust to a few poor detections. With this layout, missing corners can be predicted from the tags that were seen. With three tags, the fourth is placed to complete a parallelogram, $\mathbf{p}_4 = \mathbf{p}_1 + \mathbf{p}_3 - \mathbf{p}_2$. With two tags on the same side, the square is completed by stepping perpendicular to that side by the learned aspect ratio. With two diagonal tags, the side vector is solved from the diagonal and the aspect ratio before stepping out in the same way.

The crop region is then defined by the inner corner of each tag, i.e. the corner nearest the centre of the square region of sediment. This places the crop boundary just inside the tags, so the tags themselves are removed along with everything outside them.

6.1.3 Relating 3D points to image pixels

The tag positions are known in the 2D colour image, but the crop has to be applied to 3D points. The two are linked through the camera model. During capture the depth image is aligned to the colour image before the point cloud is exported, so every 3D point originates from one integer pixel of the colour image sensor. The cropper reads the colour camera intrinsics (focal lengths f_x,f_y, principal point (c_x, c_y) and distortion coefficients) directly from the scan's recording file and projects every 3D point back into the RGB image plane. A 3D point (X, Y, Z) in the camera frame maps to

$$u = f_x\, x_d + c_x, \qquad v = f_y\, y_d + c_y, \tag{6.1}$$

where (x_d, y_d) are the normalised coordinates $(X/Z,\ Y/Z)$ after lens distortion has been applied. The distortion is computed exactly as the RealSense SDK computes it. Using the SDK's form reprojects the cloud onto the pixel grid to within about 10^{-5} pixels.

This property also provides a self-check. If the intrinsics really belong to this cloud, every reprojected point should land almost exactly on an integer pixel. The cropper measures the root-mean-square distance of the reprojected points from the nearest integer pixel centre,

$$\rho = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left\|\mathbf{u}_i - \mathrm{round}(\mathbf{u}_i)\right\|^2}, \tag{6.2}$$

and rejects the automatic crop if ρ exceeds 0.15 pixels. The same test is used to choose how the distortion coefficients should be applied, since the stored coefficients can describe either the forward or the inverse distortion. The interpretation with the smallest ρ is kept. This guards against silently cropping a scan with the wrong camera model, for example if a recording was captured with a different resolution.

6.1.4 Building the crop volume

The simplest crop would keep every point whose pixel falls inside the tag-bounded quadrilateral. This is equivalent to cutting the cloud with a pyramid whose apex is at the camera. The walls of that pyramid follow the camera's lines of sight, so a tilted scan would be cut along a different boundary on the sediment than an untilted scan, and the scans would no longer cover the same region.

Instead, the cropper builds a vertical prism through the tag corners, so that the crop boundary is the same physical line on the bed for every scan:

1. **Fit the bed surface.** A plane is fitted to the points in a narrow band (25 pixels) just inside the crop edge. The band is used rather the whole region so that the test object or sediment relief in the middle of the square does not influence the fit. The fit uses RANSAC: planes through three randomly chosen points are tried 500 times, and the plane with the most points within 5 mm of it is kept. RANSAC is used because it ignores outliers entirely instead of averaging them in, which matters near the edge of the crop where stray points from the rig may remain.

2. **Fit the tag plane.** A second RANSAC plane is fitted to the points lying on the tags themselves. The tags can sit slightly above the sediment, so the 3D corners of the crop region are placed where the pixel rays through the inner tag corners meet this plane. For a pixel ray with direction $\mathbf{r}$ and a plane with normal $\mathbf{n}$ and offset d, the intersection is

$$\mathbf{X} = \frac{-d}{\mathbf{n}\cdot\mathbf{r}}\,\mathbf{r}. \tag{6.3}$$

3. **Extrude along the surface normal.** All points are projected onto the bed plane, and a point is kept if its projection falls inside the four 3D corners. The result is a column whose walls are perpendicular to the bed and pass through the inner tag corners, independent of the camera's viewing angle.

If no bed plane can be found, the cropper falls back to the pyramid crop described above and notes this in its output. A crop that keeps fewer than 1000 points is treated as a failure.

6.1.5 Review and manual fallback

Unless run in batch mode, each automatic crop is shown in a preview window with the removed points dimmed and the crop boundary drawn as a wireframe. The user can accept it, keep the original cloud, or switch to a manual crop. The manual crop starts from points the user picks in the viewer and then lets the box be moved and resized with the keyboard. It is also used whenever the automatic crop fails, for example when fewer than two tags were detected. This keeps the user in the loop for the cases where automatic tag detection is least reliable.

6.1.6 Outputs

For each scan the cropper writes:

- **The cropped point cloud**, in the same coordinate frame as the input. It is written with 32-bit floating-point coordinates, because Open3D's default writer produces 64-bit coordinates that the PCL-based fusion program fails to read.
- **A tag sidecar file** giving the 3D positions of the four corners of every visible tag. Cropping removes the tags from the cloud, but fusion can use them for alignment (Section 6.2.3), so their positions are measured on the uncropped cloud before they are discarded. Each tag's pixel corners are taken as the median over every frame of the recording, which removes detector jitter. A plane is fitted by least squares to the depth points on the central 70% of the tag, this central region is selected to exclude the blending of the tag and bed surface. The 3D corners are the intersections of the corner pixel rays with that plane.
- **Copies of the IMU, detection and configuration files**, so that the cropped session has the same folder layout as a raw session and can be passed to fusion unchanged.

6.1.7 Limitations

The automatic crop assumes that the sediment near the edge of the square is roughly flat, so that the band plane represents the bed. If relief reaches the crop edge, the fitted plane, and therefore the prism, can tilt. The layout learned from four-tag frames is pooled across all poses, although perspective changes with tilt, so corners extrapolated from only two tags in a strongly tilted view may be slightly misplaced. The prism also has no height limit, so isolated noisy points directly above or below the square are kept.

6.2 Multi-Scan Fusion

Together is Almost Always Better

A single scan is noisy, has holes where relief hides the surface, and has errors that depend on the viewpoint. Combining registered scans addresses all three. Views from different directions see regions occluded in others, and the mean of n measurements with independent errors of standard deviation σ has a standard deviation of $\sigma / \sqrt{n}$. This however does not alleviate errors shared by all scans, such as a calibration or scale bias, all of which cannot be reduced through averaging.

Fusion is performed by a C++ program, `fuse_scans`. It runs as a sequence of stages: load and preprocess, estimate initial poses from the rig, coarse alignment, fine registration and reconciliation. The program can stop after coarse and after fine alignment to show every scan in an interactive viewer, colour-coded and listed with its alignment quality. From there the user can exclude a bad scan, nudge one into place by hand, or rerun a stage, before the session is fused. In batch processing these reviews are skipped.

The common frame is that of the first untilted scan, called the *reference*. The remaining scans are processed in order of increasing tilt, so that each new scan is as similar as possible to what has already been placed.

6.2.1 Preprocessing

Each cropped cloud is first thinned with a voxel grid of user defined size, default of 3 mm: space is divided into 3 mm cubes and all points in a cube are replaced by their centroid. This makes the density uniform, which most of the later algorithms assume, and it greatly reduces their cost. It also sets the finest detail the fused result can contain, so the voxel size is chosen close to the noise floor of the sensor. A statistical outlier filter then removes isolated points, defined as points whose mean distance to their 30 nearest neighbours is more than one standard deviation above the average.

Finally, a surface normal is estimated at every point from the principal axes of its neighbourhood within 15 mm. All normals are oriented to point towards the camera. This matters later: normals from several scans are averaged, and a normal and its opposite would cancel. The normals are computed once here and carried through every later transformation, because re-estimating them inside registration accounted for most of the program's run time.

6.2.2 Initial poses from the rig and IMU

Registration algorithms only converge when they start close to the right answer. The rig provides a natural starting point, because each scan's folder name records the commanded rotation and tilt. These commanded angles proved to be inaccurate, however. Commanded tilts of 15° produced measured tilts between 13.6° and 17.4°, and individual steps varied from 0.6 to 1.24 times the commanded step. In addition, because the rotating mount sits on top of the tilt stage, the tilt axis appears in a different direction in the camera frame for every rotation position, so a single fixed tilt axis cannot describe the rig.

The tilt is therefore *measured* rather than assumed. When the camera is stationary its accelerometer reads only gravity. The rotation that carries the gravity vector of a scan onto the gravity vector of the flattest scan at the same rotation position is exactly the tilt the mount applied, because the hinge axis is horizontal. Gravity readings with a magnitude outside 9.0–10.6 m/s^2 are rejected, since they indicate the camera was moving.

Two further quantities cannot be read from the IMU and are found by search:

- **The IMU-to-cloud axis convention.** The accelerometer and the exported point cloud use different axis conventions. Four candidate conventions (the identity and a 180° flip about each axis) are tried. For each, a few low-tilt scans are placed using the IMU tilt and compared with the reference scan. The convention with the best agreement is kept. On real data the correct convention gave residuals of 7–18 mm, against 42–76 mm for the wrong one.
- **The tilt pivot.** The IMU gives the angle of tilt but not the point the camera tilts about. The pivot is found by a coarse-to-fine grid search over a 0.12 m range, again scoring how well the placed scans agree with the reference. The convention and the pivot are searched together because an error in either can mask an error in the other.

The rotation direcition poses a different problem. Since the rotation stage is performed manually, the capture

code names each rotation as a positive 45°increment, regardless of direction or order. IMU data cannot resolve this issue since the rotation axis is parallel to the gravity axis, and therefore can not be estimated using accelerometer data. The program therefore tries every combination of clockwise and anticlockwise steps, places one representative scan per rotation position accordingly, and keeps the combination under which the scans agree best. A joint search is used rather than deciding each step on its own, because a wrong early step makes all later steps fit worse, which only a joint search can detect. When AprilTag alignment is enabled, the tags measure the rotation angle of each rotation position directly, and the search is needed only for positions where no tags were seen.

6.2.3 Coarse alignment

The coarse stage places each scan to within a few millimetres of its final position. Scans are added one at a time, and each is aligned against a map built from every scan placed so far, rather than against only the previous scan. This reduces the accumulation of error (drift) along the sequence. Up to three candidate poses, or hypotheses, are generated for each scan:

1. **The rig seed**: the pose predicted by the measured tilt and inferred rotation of Section 6.2.2
2. **The AprilTag pose**, if tag alignment is enabled. Each tag contributes four 3D corners from the sidecar file, which are matched against the same tag corners already placed in the common frame. The best rigid transformation between the matched corners is found in closed form with the method of Umeyama [**?**], which uses a singular value decomposition of the corners' cross-covariance matrix. No scale term is estimated, because both point sets are already in metres. When gravity is available, a constrained version is used instead: the tilt is taken from the IMU and only the turn about the vertical axis and the translation are solved from the tags. The tags a scan sees often lie nearly in a line, so the tilt about that line would otherwise rest on depth noise across a single 80 mm tag, whereas the IMU measures tilt well. On one test session this reduced the disagreement between tag poses and the final registered poses from 1–2.5° to 0.2–0.6°.
3. **A feature-based pose**, computed only if the rig seed overlaps the map poorly. Fast Point Feature Histograms (FPFH) [**?**] describe the local shape around every point. Points with similar descriptors in the scan and the map are paired, and a RANSAC variant [**?**] repeatedly picks three pairs, rejects them early if the triangles they form have different shapes, and keeps the transformation that brings the most points into agreement. This method needs no initial guess, but it is unreliable on smooth, nearly flat sediment with few distinctive shapes, which is why it is only a fallback.

To choose between hypotheses, each is scored by comparing the transformed scan with the map. A sample of scan points is matched to the nearest map point. Points within 10 mm are counted as overlapping, and the score is the root-mean-square distance of the overlapping points divided by the fraction of points that overlap, and the lowest resulting value is chosen. Dividing by the overlap fraction prevents a hypothesis from scoring well by fitting a small sliver of the scan very tightly.

For nearly flat scenes such as the sediment bed, this nearest-neighbour score has a blind spot. If a scan is slid or spun within the plane of the bed, every point still has a nearby neighbour on the flat map, so the score hardly changes. In that case the program instead compares heights. Both clouds are expressed in the frame of the bed plane, binned onto a 10 mm grid, and the score is computed from the differences in mean height between corresponding cells. A 5 mm in-plane slide raised this height-field score by 36%, but the nearest-neighbour score by only 16%. Whether a scene counts as flat is decided once from the reference scan.

The rig seed is accepted directly if at least 15% of the scan overlaps the map. Only if it fails is FPFH tried, and the better of the two is kept. A tag hypothesis competes with the result on the same score. If the tag pose and the geometric pose disagree by more than 10°, a warning is printed, because a disagreement that large means that one of them is wrong, not merely imprecise.

6.2.4 Fine registration

The fine stage refines each coarse pose with the Iterative Closest Point (ICP) algorithm [**?**]. ICP alternates between pairing every scan point with its nearest map point and solving for the rigid transformation that minimises the distance between the pairs, until the pose stops changing. The point-to-plane form of ICP [**?**] is used, which minimises the distance of each scan point from the tangent plane at its paired map point rather than from the map point itself:

$$E(\mathbf{R}, \mathbf{t}) = \sum_{(i,j)} \Big((\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_j) \cdot \mathbf{n}_j \Big)^2, \tag{6.4}$$

where $\mathbf{p}_i$ is a scan point, $\mathbf{q}_j$ its paired map point, $\mathbf{n}_j$ the map normal, and $\mathbf{R}$, $\mathbf{t}$ the rotation and translation being solved for. Because of the voxel grid, a scan point and its true counterpart on the map are generally not the same physical point, only points on the same surface. Point-to-plane ICP lets points slide along the surface instead of penalising this offset, so it converges faster and more accurately on continuous surfaces than point-to-point ICP. Each step is solved with the linear least-squares approximation of Low [**?**]. Pairs further apart than 20 mm are ignored, so a coarse pose must be within roughly this distance for ICP to recover it, and at most 60 iterations are run.

As in the coarse stage, each scan is registered against a map of all scans placed so far. The map is stored as a voxel grid in which each voxel keeps running sums of the positions and normals of every scan that falls into it. Every scan therefore carries equal weight in the map. An earlier version re-thinned the merged map after every scan, which gave the newest scan about half the weight wherever it overlapped, so the map followed the latest scan rather than the consensus.

After the stage, a scan is flagged for review if ICP failed to converge, hit its iteration limit, or scored more than 1.4 times worse than the median scan of the session. The threshold is relative because what counts as a good score depends on the voxel size, the sensor noise and the relief of the surface. A scan whose refinement fails keeps its coarse pose, so that it remains visible in the review instead of silently disappearing.

6.2.5 Reconciling the overlapping scans

Once registered, the scans could simply be combined into one cloud and thinned with a voxel grid. This was the original approach, and it gives a poor result. The remaining registration error (a few millimetres) is larger than the output voxel, so where scans overlap, their surfaces land in neighbouring voxels and survive as two or three slightly offset sheets. A statistical outlier filter cannot remove a duplicate sheet, because the duplicate is just as dense as the real surface. The fusion step therefore reconciles the scans.

Weighting each measurement. Not all measurements are equally reliable, so each point i is given a weight

$$w_i = c_{s(i)} \cdot \max(0.05,\ |\hat{\mathbf{n}}_i \cdot \hat{\mathbf{v}}_i|) \cdot \min\left(1,\ \frac{r_0^2}{r_i^2}\right). \tag{6.5}$$

The first factor, $c_{s(i)}$, is the registration confidence of the scan the point came from, based on how much of the scan overlapped the map and how closely. The second factor is the cosine of the angle between the surface normal $\hat{\mathbf{n}}_i$ and the direction to the camera $\hat{\mathbf{v}}_i$. This factor is included as stereo depth error rises sharply when the surface is seen at a grazing angle, so such points are down-weighted. The third factor reflects the range r_i from the camera. This factor is included as the depth error of a stereo camera generally grows with the square of the range, so points beyond the reference range r_0 are down-weighted accordingly.

Grouping measurements into cells. For a nearly flat scene the points are expressed in the frame of the bed plane and binned into the user defined sized cells in the plane, each holding the heights of all points that fall into it. This *height-field* mode is chosen automatically when the scene is flat, which applies to the sediment bed. Scenes with strong relief use a volumetric mode instead, in which points are binned into 3D voxels and reconciled along the local surface normal.

Choosing one value per cell. Within each cell, the measurements are combined with a *trimmed weighted mean*. The samples are sorted by height, and the lowest and highest 25% of the total weight are discarded before the weighted mean of the remainder is taken. A plain weighted mean would let a single misregistered scan drag the surface towards it, while a weighted median discards most of the information. The trimmed mean lies between the two. Trimming by weight rather than by count means that a single high-confidence measurement cannot be outvoted by several poor ones. On a synthetic test with one of four scans misregistered by 8 mm, the error of the fused surface was 2.00 mm with a plain weighted mean and 0.35 mm with the trimmed mean. Overall, reconciliation halved the error of the original merge-and-thin approach, from 0.96 mm to 0.45 mm RMS.

Removing off-surface outliers. Points that still lie off the surface are then removed. For each point a plane is fitted to its 31 nearest neighbours, and the point is discarded if its distance from that plane is more than three standard deviations above the average. This replaced the statistical outlier filter used earlier, which judges points by local density. That filter removed genuine but sparsely sampled parts of the surface and punched holes in the fused result, whereas the plane test only removes points that are actually off the surface.

Smoothing. Finally, the surface is smoothed with Moving Least Squares (MLS) [**? ?**]. Around each point, MLS fits a quadratic surface to the neighbours within 10 mm and moves the point onto it, which also yields a clean normal. Because the fitting radius is larger than the residual misregistration, this removes the last traces of offset sheets and sensor noise along the surface normal while keeping features larger than the radius. It is, however, a low-pass filter: relief with a wavelength comparable to or smaller than the 10 mm radius is attenuated. This must be kept in mind when interpreting the small-scale roughness parameters of Section 6.3. [TODO: State whether the results in Ch. X were computed with or without MLS, and consider a short sensitivity check with `--no-mls`.]

The result is a single point cloud per session, in the frame of the reference scan, with one point per occupied cell and a normal at every point. The program also reports the thickness of the overlapping surface, i.e. the spread of heights within a cell, before and after reconciliation.

6.2.6 Limitations

Registration is sequential: each scan is aligned against the scans before it, and no global optimisation revisits earlier poses afterwards, so an early error can propagate. The rig seed is accepted whenever it overlaps the map by 15% or more. On a flat bed even a pose with a wrong rotation can overlap that much, in which case only the tag hypothesis can correct it, so tag alignment should be enabled for sediment scans. The pivot of the rotation stage is not estimated, so an offset between the turntable axis and the camera must be absorbed by the tags or by ICP. The height-field mode assumes the surface has a single height at every point, so overhangs and vertical faces, such as the inside walls of a burrow, are averaged away. Finally, whilst the smoothing stage helps to remove sensor noise and point offsets, it does effectively low-pass filter the surface, and thus can remove sharp edges in the scan.

6.3 Surface Roughness Estimation

[TODO: review this!]

The roughness program, `roughness_scans.py`, takes a fused point cloud and describes the roughness of the surface it represents. It runs five steps: level the scan, crop it, remove large-scale trends, resample it onto a regular grid, and compute the metrics. Most of the metrics follow the areal surface-texture parameters of ISO 25178-2 [**?**], so that results can be compared with other surface-metrology studies. Internally all units are in metres with reported units being millimetres.

6.3.1 Levelling

The fused scan is in the frame of the reference camera, so the sediment bed is generally tilted relative to the coordinate axes. Roughness parameters measure heights relative to a mean surface, so the scan is first levelled. This involves rotating scan so that the mean plane of the bed is horizontal and height is measured along the z axis. The mean plane is found by principal component analysis. The plane passes through the centroid of the points, and its normal is the direction in which the points vary least. To prevent large features from tilting the plane, the fit is repeated with sigma clipping, where points further than three times the spread estimate are dropped and the plane is refitted, this is then repeated up to five times. The spread is estimated with the median absolute deviation (MAD), scaled by 1.4826 so that it equals the standard deviation for normally distributed data. The reason for using MAD instead of normal standard deviation, is that MAD is less susceptible to the changes that result from discarded samples.

The levelled frame has its origin at the centroid, $+z$ along the plane normal pointing towards the camera, and x along the camera's original x axis projected onto the plane. Keeping the camera's horizontal direction means that the direction of ripples or other features reads the same in the results as in the images. The scan can then optionally be cropped to a region of interest or trimmed by a fixed margin around its border, where fused data are usually the sparsest.

6.3.2 Detrending

Although the surface has been levelled, there are still some sediment features forms that are not related to the roughness of said sediment. For example, a concave section of sediment, or residual tilt from the levelling stage. Removing these forms from the surface results in the isolated sediment texture surface, called the residual. For detrending, these non-textural forms are modelled as a 3D polynomial surface:

$$p(x, y) = \sum_{i+j \le n} a_{ij} x^i y^j, \tag{6.6}$$

of degree n from 1 (planar) to 4 (quartic), and the residual $z - p(x, y)$ is kept as the roughness surface. The coordinates are scaled to the range $[-1, 1]$ before fitting so that the fit remains numerically stable at higher degrees, and the same sigma clipping as in levelling stops ripple crests, troughs or burrow mounds from dragging the trend.

The choice of degree is a trade-off. A low degree may leave some non-texture form in the residual and overestimate roughness. A high degree may capture more complex forms but can also remove features of interest. A polynomial of degree n can follow roughly $n/2$ wavelengths, so on a small patch it can remove relief that is part of the texture being measured. The program therefore computes the metrics for every degree requested, using the same levelling, crop and grid for each so that the results are directly comparable, and raises a warning if a higher-order fit removes much more height variance than the planar fit while the patch spans fewer than n dominant wavelengths. [TODO: State which detrend degree is used for the reported results and why.]

6.3.3 Gridding

Most roughness parameters are defined on a regular grid of heights, so the residual surface is resampled onto a square grid. By default the cell size is twice the median spacing between neighbouring points, so that most cells contain several points. The height of a cell is the mean of its points. Averaging c points with independent noise of standard deviation σ reduces the noise to $\sigma / \sqrt{c}$, so the gridding also acts as a mild low-pass filter at the scale of one cell.

Some cells contain no points, therefore any small enclosed holes, no larger than two cells across, are filled by linear interpolation from the surrounding cells. Larger holes and gaps at the border are left empty and excluded from every metric, because interpolating across them would result in inaccurate metrics..

6.3.4 Height parameters

With z_k being the gridded heights relative to their mean and N being the number of valid cells, the height parameters are:

$$S_a = \frac{1}{N} \sum_k |z_k|, \qquad S_q = \sqrt{\frac{1}{N} \sum_k z_k^2}, \tag{6.7}$$

$$S_{sk} = \frac{1}{S_q^3} \frac{1}{N} \sum_k z_k^3, \qquad S_{ku} = \frac{1}{S_q^4} \frac{1}{N} \sum_k z_k^4, \tag{6.8}$$

together with the highest peak S_p, the deepest valley S_v and the total height $S_z = S_p + S_v$. S_a and S_q are the arithmetic and root-mean-square heights and are the most common single measures of roughness. The skewness S_{sk} indicates whether the surface is dominated by peaks ($S_{sk} > 0$) or by pits ($S_{sk} < 0$). A bed riddled with burrow openings would be expected to have negative skewness. The kurtosis S_{ku} indicates how sharp the features are, a surface with normally distributed heights has $S_{ku} = 3$, while isolated sharp peaks or pits raise it.

6.3.5 Rugosity and slope

Rugosity is the ratio of the true area of a surface to the area of its projection onto the horizontal plane. It is the areal equivalent of the chain-and-tape method used in benthic ecology, in which a chain is laid along the contours of the bed and its length compared with the straight-line distance. A perfectly flat surface has a rugosity of exactly 1, and any relief raises it.

To compute the true area, each square of four neighbouring grid cells is split along a diagonal into two triangles, and the area of each triangle is computed from the cross product of its edges. Summing over all squares whose corners are valid gives

$$\text{rugosity} = \frac{\sum A_{\text{triangles}}}{N_q \, g^2}, \tag{6.9}$$

where N_q is the number of valid squares and g the grid spacing. The ISO 25178 developed interfacial area ratio is the same quantity expressed as a percentage excess, $S_{dr} = (\text{rugosity} - 1) \times 100\%$. The root-mean-square slope of the surface, S_{dq}, is computed from the same squares.

Rugosity, S_{dr} and S_{dq} all depend strongly on the grid spacing. Measurement noise adds small random slopes to every cell, and the finer the grid, the steeper those slopes become, so these parameters grow as the grid is refined even if the surface is unchanged. Results computed with different grid spacings are therefore not comparable, and all scans in this work are processed with the same spacing. [TODO: state the value used.]

6.3.6 Spatial and spectral parameters

The height parameters describe how tall features are, but not how large or how far apart they are. Two further groups of parameters capture this.

Autocorrelation. The autocorrelation function measures how similar the surface is to a copy of itself shifted horizontally. It equals 1 at zero shift and decays as the shift grows. The autocorrelation length S_{al} is the shortest shift at which it falls below 0.2, and gives the typical horizontal size of the features. The texture aspect ratio S_{tr} is the ratio of the shortest to the longest such distance over all directions: it is close to 1 for a surface that looks the same in every direction, such as a randomly burrowed bed, and close to 0 for a strongly directional one, such as a rippled bed. The autocorrelation is computed with the fast Fourier transform, counting only pairs of valid cells.

Power spectrum. The power spectral density (PSD) breaks the surface down into sinusoidal components and shows how much height variance each wavelength contributes. A Hann window is applied before the transform to reduce artefacts from the edges of the patch. The strongest peak of the PSD gives the dominant wavelength and direction of any regular pattern, such as ripples, and the texture direction S_{td} is reported as the direction of the crests. [TODO: Note: S_{td} here is the crest direction, i.e. 90° from the peak of the angular spectrum; check this against the convention of any software you compare with.] Many natural surfaces are *self-affine*: their PSD follows a power law $C(q) \propto q^{-2(1+H)}$ in wavenumber q over a wide range of scales [**?**]. Fitting a straight line to the log-log PSD gives the Hurst exponent H, which describes how roughness changes with scale, and the fractal dimension $D_f = 3 - H$. The fit is restricted to wavelengths between four grid cells, below which gridding distorts the spectrum, and a quarter of the patch width, above which there are too few repetitions to estimate the power reliably.

6.3.7 Additional measures and uncertainty

Several further outputs support the interpretation of the main parameters:

- **Multi-scale roughness.** S_q and rugosity are computed in square windows of increasing size, each with its own plane removed, to show how roughness depends on the scale of observation.
- **Confidence intervals.** 95% confidence intervals for S_q, S_{dr} and S_{al} are estimated with a block bootstrap: the surface is divided into square windows, which are resampled with replacement 1000 times. Neighbouring windows are correlated, so these intervals are likely to be somewhat optimistic.
- **Terrain ruggedness.** The Terrain Ruggedness Index [**?**] and the Vector Ruggedness Measure [**?**], widely used for seabed and landscape habitat analysis, are included for comparison with the ecological literature.
- **Noise correction.** Given an estimate of the sensor noise,either as a per-point standard deviation or from a scan of a flat plate processed in the same way, the noise contribution is subtracted from S_q and S_{dq}, assuming the noise adds independently to the true surface. A warning is raised if noise accounts for more than half of the measured slope, since rugosity then mainly measures noise.

All random choices use a fixed seed, so repeated runs give identical results, and every run records the hash of the code and of the input scan.

6.3.8 Validation

The program was checked in two ways. A test suite generates synthetic surfaces with known answers, such as sinusoids, for which S_q, S_a, the wavelength and even the exact rugosity follow in closed form, and random fractal surfaces with a prescribed Hurst exponent. Each metric is confirmed to recover the expected value, including when trends, noise, holes and a tilted camera are added. Secondly, a scan of a test object can be compared directly with its analytic definition: the reference surface is sampled, passed through exactly the same levelling, detrending and gridding, and the two sets of metrics are compared. This measures the error of the complete capture, fusion and roughness chain, and is used in Chapter **??** [TODO: check label].

6.3.9 Limitations

The parameters follow ISO 25178-2 definitions but are not fully ISO-conformant. The standard defines them on a surface that has been band-limited with Gaussian filters, whereas here the band is set implicitly by the grid cell, the MLS smoothing in fusion, the polynomial degree and the size of the patch. The values are therefore directly comparable between scans processed in this pipeline with the same settings, but should be compared with values from commercial surface-metrology software with care. The surface is also treated as a single-valued height map, so overhangs are lost.

6.4 Feature Detection

The roughness parameters describe the surface as a whole, but the biological question also requires counts of discrete features, in particular the burrow openings created by sandprawns. The final stage of the pipeline is intended to detect these automatically from the depth and colour images.

[TODO: Not yet implemented.]

Chapter 7

Experimental Validation

In order to validate the processing pipeline as a whole, two methods are used. Firstly the pipeline is applied to experimental data captured during the characterisation stage of the known relief objects. Whilst this does not acurately describe the sediment surface in situ, it allows for an analytical evaluation the pipeline, whilst providing some resemblance for the true conditions.

Secondly, scans of the surface of sand, taken from the langebaan lagoon, was performed. This was taken for dry, wet and submerged sand, in both flat and shaped forms. This experimentation provides little analytical results, but helps to get a qualitative understanding of the performance in conditions closer to environmental.

7.1 Known Relief and Controlled Targets

7.1.1 Process

The rugosity estimation pipeline is applied to scans taken of reference objects shown in chapter C. These scans are re-used from the characterisation process chapter 5. These scans had the filters identified in chapter 5 applied to them. Cropping and fusion was then applied. This reconstructed surface was then cropped down to just the reference object, and the evaluation was performed on it. The results of which are then compared to the results obtained via analytical evaluation of the reference object file .stl. This provides an analytical comparison between the results of scanning and processing, and the known relief.

7.1.2 Results

Each smooth shape (big egg box and small egg box) were processed using the rugosity_metrics processing pipeline compared against their respective reference .stl files. The evaluation was performed on both fused and single scan results, and each metric applied all four detrending methods (planar, quadratic, cubic and quartic). These rugosity metrics are shown in Table 7.1 and Table 7.2, which are in turn plotted in Figure 7.1 and Figure 7.2.

There are two main points to be discussed with these results. Firstly is the effects of the different detrending results. The results show that planar and quadratic detrending methods tend to produce results that better match the ground truth. Care must be taken here since this analysis applies the same detrend to the ground truth. Thus the planar detrending should be the best representation of the ground truth rugosity. The primary reason for planar and quadraticdetrending giving the best performance can be seen in Figure 7.1, where the lower wavelength shape closely alignes with the higher order detrending methods, and thus they tend to remove this information. These can be visually seen in chapter F where the cubic and quartic detrending methods tend to remove actual shape of the big egg box surface, rather than underlying trends.

Secondly, and most importantly, is that these results are innacuratly representing the truth. The plots show that

single scans generally produce rugosity metrics that are more closely aligned with the reernce objects. This is contrary to what is expected, as the fused results generally provide a closer representation to the ground truth. The fundamental reason for this discrepency is actually a misleading result. The reconstrucitons for these reference objects are not completely accurate, and all feature some scale error, and underrepresetnt the amplitude of the reference waveforms. This means that the reconstructions will always report a lower rugosity compare to what the true waveforms have. This is where the inflated single scan performance arives. Since the single scans have a higher depth noise, this noise arteficially increases the roughness of the surface. This can be seen in Figure 5.10a and Figure 5.9a. This arteficial increase in the roughness, results in a higher rugosity, which results in an arteficially better result.

Table 7.1: Each value is over n = 1 rows.

Detrend Method	Scan	Rugosity mean ± SD	Ref rugosity mean ± SD
cubic	Big Egg Box Fused	1.19	1.27
	Big Egg Box Single	1.26	1.26
planar	Big Egg Box Fused	1.25	1.34
	Big Egg Box Single	1.31	1.34
quadratic	Big Egg Box Fused	1.25	1.34
	Big Egg Box Single	1.31	1.34
quartic	Big Egg Box Fused	1.19	1.27
	Big Egg Box Single	1.26	1.26

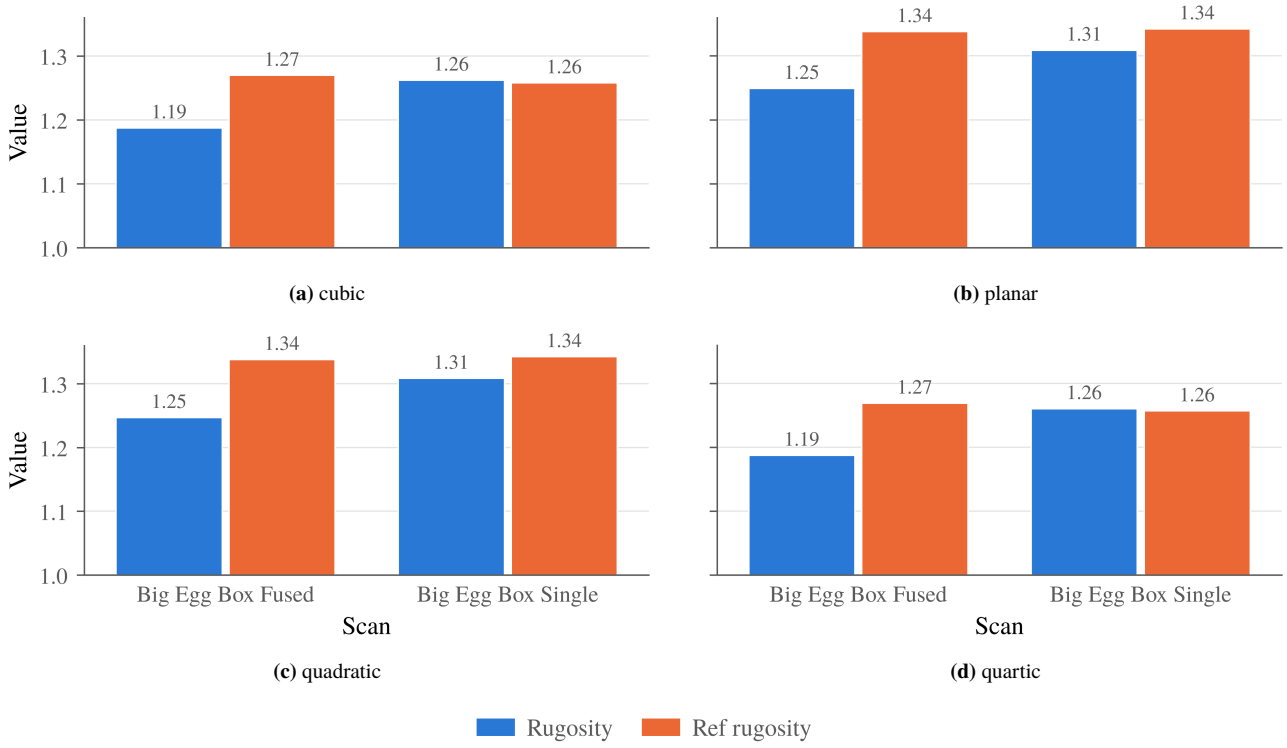
Table 7.2: Each value is over n = 1 rows. Fixed: Grid nx = 45.

Detrend Method	Scan	Rugosity mean ± SD	Ref rugosity mean ± SD
cubic	Small Egg Box Fused	1.26	1.69
	Small Egg Box Single	1.31	1.69
planar	Small Egg Box Fused	1.26	1.69
	Small Egg Box Single	1.31	1.69
quadratic	Small Egg Box Fused	1.26	1.69
	Small Egg Box Single	1.31	1.69
quartic	Small Egg Box Fused	1.25	1.69
	Small Egg Box Single	1.31	1.69

7.2 Controlled Laboratory Sand

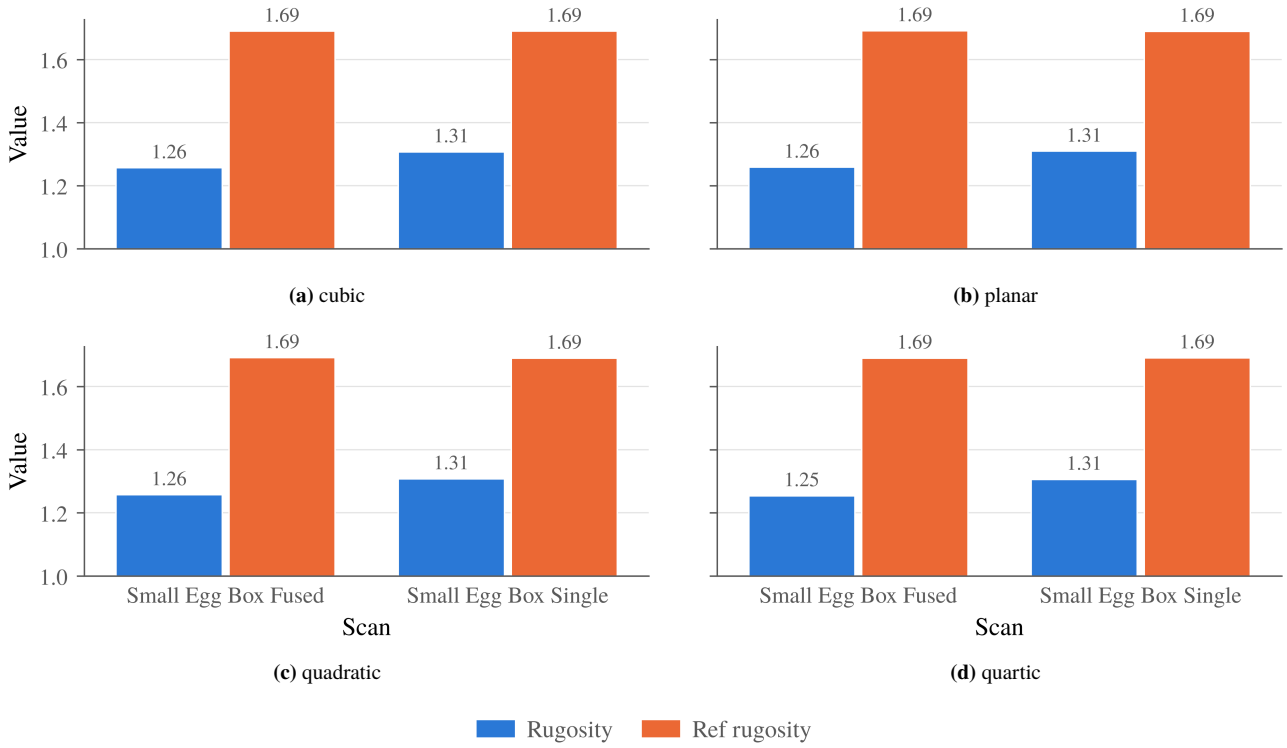
Whilst this reference object comparison provides an analytical evaluation, it does not align with real world conditions. Due to time constraints, real world implementation was not able to be performed, however a more realistic scan was taken, using sand collected from the langebaan lagoon. Approximately 13 kg of sand was collected dry, and was placed in a a 10 l plastic container. This was then flattened to produce a surface that should give rugosity metrics of 1. next, the sand was indented with the known reference shapes, and scans were taken. Next the sand was wetted with 2 kg of water. This approximates the wet intertidal sediment. The sand was then flattened again for scans, and then indented with the known reference shapes.

Finally, for purely qualitative performance, slightly more water was added, enough to produce pools of water on the surface. The sand was then shaped by hand to have some approximate natural texture. This was then scanned. This mainly gives an estimation of the performance on more water logged regions of the sediment, and also gives an estimate on what rugosity measurements would look.



(a) cubic **(b)** planar **(c)** quadratic **(d)** quartic

Figure 7.1: Mean ± SD of n = 1 rows per bar.



(a) cubic **(b)** planar **(c)** quadratic **(d)** quartic

Figure 7.2: Mean ± SD of n = 1 rows per bar. Fixed: Grid nx = 45.

7.2.1 Results

These results show a much better trend in the performance of the system. As can be seen in the dry flat analysis, the fused scan results in a rugosity of [TODO: Insert metric], which is lower than that of the unfused scan which gives a rugosity of [TODO: give metric]. This is primarily due to the fact that single scans have a higher depth noise, and on flat planar surfaces, this raises the rugosity beyond the actual natural sediment. This is what is expected, and provdes a better representation than the referenc object validation.

Unfortunately these results dont have ground truth, since they were taken from sediment that was manually flattened and shaped. However te results do show that planar surfaces result in rugosirt that is far closer to 1 than shaped surfaces, which is expected since a flat surface would result in a rugosity of exactly 1.

An interesesting results can be seen with the soaked scan. This scan is the most representative of what sediment in situ might represetnt, and the scans show that system is able to accurately represent sediment shape, even when partially submerged by water. Providing useful insigt to show the robusteness to saturated sediment.

antoher intereting metric can be seen with the wet planar surface. In the midle of this surface is a large region of high noise. This is atributed to high specular reflections of the ir projector on the predominatly flat surface, with a water film. This shows that performance can degrade under such dconditions where the projector pattern can be reflected directly back into the ir sensors. Therefore this shows that care must be taken to prevent such specular reflections.

Table 7.3: Each value is over n = 4 rows. Averaged over: Detrend (4), Primary (2), Flags (3), Str bounded (2), Detrend n coef (4), Detrend order (4) and 1 more.

Label	Rugosity mean ± SD
Dry Flat	1.00340 ± 0.00021
Dry Flat Single	1.01933 ± 0.00018
Dry Shaped	1.05806 ± 0.00037
Dry Shaped Single	1.07912 ± 0.00036
Soaked	1.02659 ± 0.00104
Soaked Single	1.02922 ± 0.00077
Wet Flat	1.00931 ± 0.00013
Wet Flat Single	1.02429 ± 0.00014
Wet Shaped	1.05924 ± 0.00023
Wet Shaped Single	1.08549 ± 0.00011

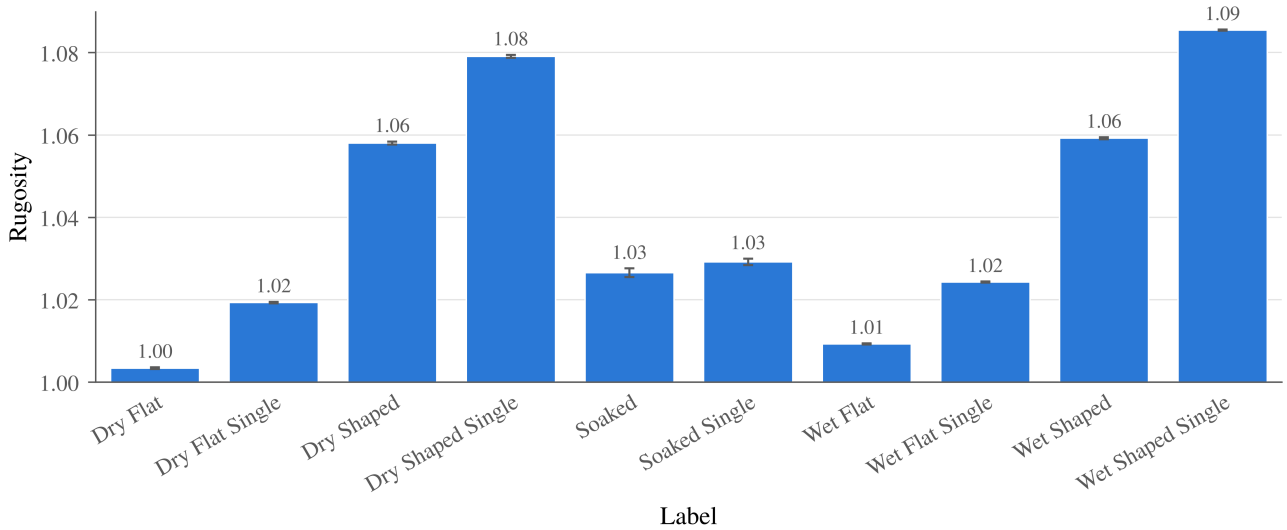


Figure 7.3: Mean ± SD of n = 4 rows per bar. Averaged over: Detrend (4), Primary (2), Flags (3), Str bounded (2), Detrend n coef (4), Detrend order (4) and 1 more.

Chapter 8

Discussion

8.1 Optimal Configuration

8.2 Roughness Estimation

8.3 Feature Detection

Chapter 9

Conclusions

Chapter 10

Further work

10.1 Characterisation

10.2 Hardware Choices

Appendix A

GitHub repository

This appendix provides a link to the GitHub repository where the code and additional materials for this final year project report can be found.

A.1 Title for appendix A.1

- GitHub Repository: Repo name

Appendix B

Theory

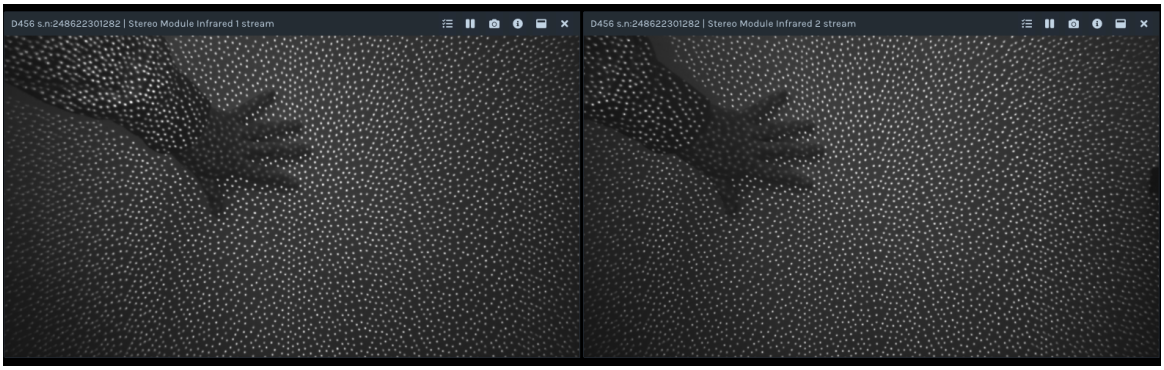


Figure B.1: D456 Semi-Random IR Dot Pattern Projected Onto a White Wall, With a Hand

Appendix C

Reference Objects

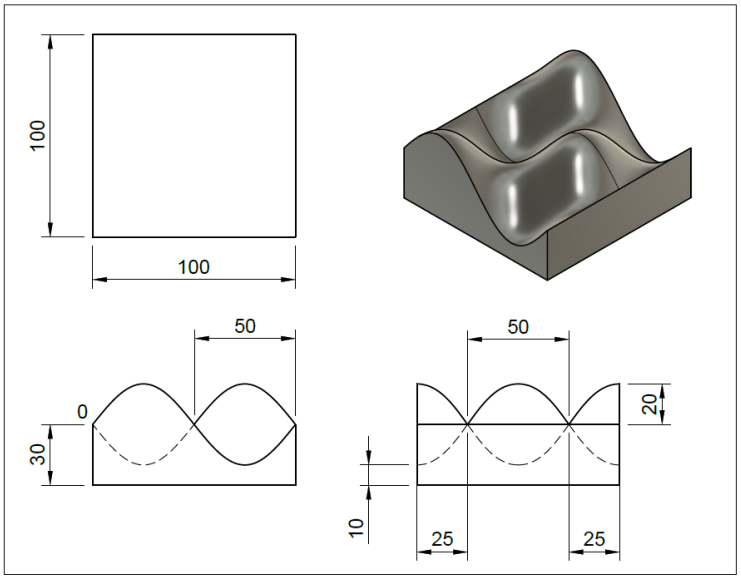


Figure C.1: Large Egg Box Surface Reference Object.

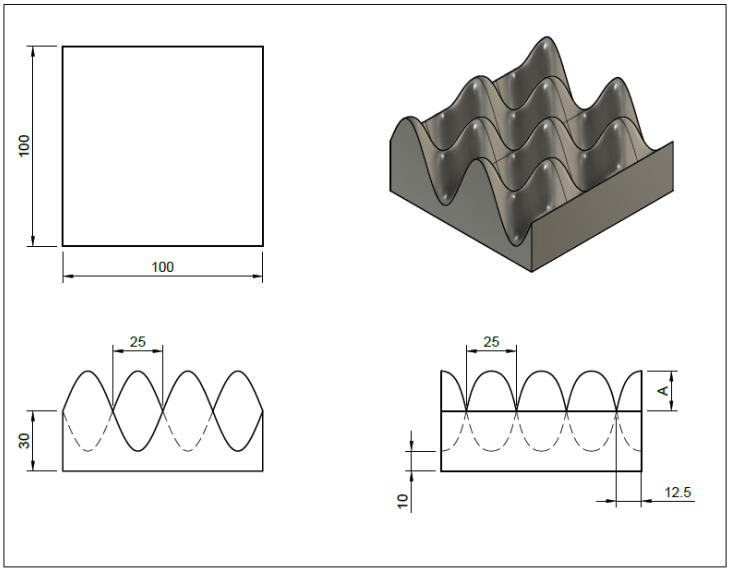


Figure C.2: Small Egg Box Surface Reference Object, Printed with amplitude A = 15mm.

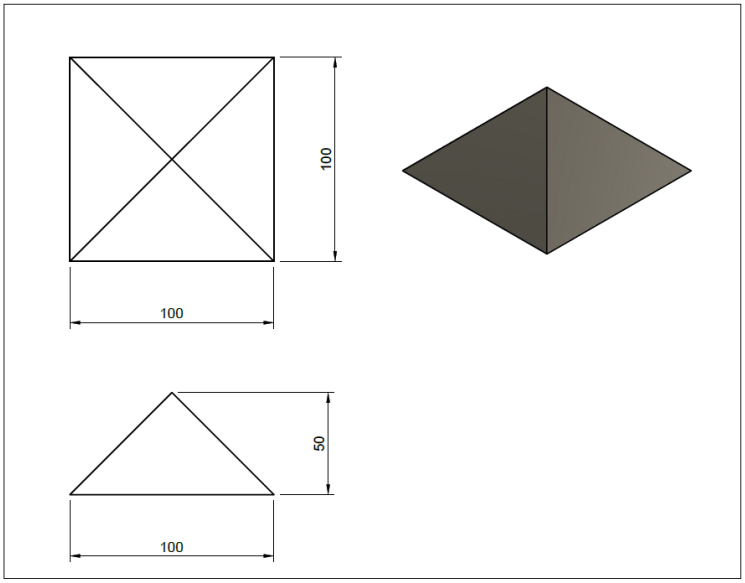


Figure C.3: Large Pyramid Surface Reference Object.

Appendix D

Characterisation Results

D.1 Planar Characterisation Results

Table D.1: Results for Planar Characterisation Against Distance. Per-Trial Values Over n = 385-552 Frames.

Distance along optical axis (mm)	Median depth error (mm) per-trial median [Q1, Q3]	Depth RMSE (mm) per-trial median [Q1, Q3]	Plane-fit RMS (mm) per-trial median [Q1, Q3]	Valid pixels (%) per-trial median [Q1, Q3]
604	4.70 [4.70, 4.70]	4.90 [4.84, 4.96]	1.22 [1.21, 1.25]	99.6 [99.6, 99.7]
628	4.60 [4.60, 4.80]	4.86 [4.81, 4.90]	1.29 [1.27, 1.31]	99.5 [99.5, 99.6]
667	6.40 [6.40, 6.60]	6.64 [6.58, 6.70]	1.56 [1.53, 1.59]	99.5 [99.4, 99.5]
723	5.60 [5.30, 5.60]	6.13 [6.04, 6.23]	2.64 [2.57, 2.70]	97.7 [97.5, 98.0]
760	6.10 [5.80, 6.10]	6.66 [6.54, 6.76]	2.81 [2.73, 2.88]	97.7 [97.5, 97.9]
821	7.80 [7.50, 7.80]	8.35 [8.21, 8.47]	3.27 [3.18, 3.35]	97.4 [97.1, 97.6]
880	7.30 [6.90, 7.30]	8.37 [8.21, 8.54]	4.05 [3.95, 4.17]	96.9 [96.6, 97.2]
903	8.10 [7.70, 8.10]	9.41 [9.20, 9.62]	4.58 [4.45, 4.70]	96.6 [96.2, 96.9]

Table D.2: Results for Planar Characterisation Against Scan Angle. Per-trial values over n=527-574 frames.

AbsMean IMU angle	Axis	Median depth error (mm) per-trial median [Q1, Q3]	Depth RMSE (mm) per-trial median [Q1, Q3]	Plane-fit RMS (mm) per-trial median [Q1, Q3]	Valid pixels (%) per-trial median [Q1, Q3]
1.7	pitch	5.0 [5.0, 5.1]	5.4 [5.3, 5.5]	1.62 [1.57, 1.66]	99.4 [99.3, 99.5]
2.5	roll	5.8 [5.7, 5.9]	6.2 [6.1, 6.3]	1.69 [1.65, 1.74]	99.2 [99.1, 99.3]
2.8	pitch	6.2 [6.1, 6.3]	6.8 [6.7, 6.8]	1.60 [1.55, 1.65]	99.5 [99.4, 99.6]
4.4	pitch	5.9 [5.8, 6.0]	6.1 [6.0, 6.2]	1.59 [1.55, 1.64]	99.4 [99.3, 99.5]
5.0	roll	5.5 [5.4, 5.6]	5.9 [5.8, 6.0]	1.67 [1.62, 1.72]	99.2 [99.1, 99.4]
6.2	roll	6.5 [6.4, 6.6]	7.0 [6.9, 7.1]	1.69 [1.64, 1.73]	99.2 [99.0, 99.3]
7.0	pitch	6.1 [6.0, 6.2]	6.4 [6.3, 6.4]	1.63 [1.59, 1.68]	99.5 [99.3, 99.6]
8.8	roll	9.8 [9.7, 9.9]	10.9 [10.8, 11.0]	1.72 [1.67, 1.77]	99.2 [99.1, 99.4]
9.3	pitch	6.7 [6.6, 6.8]	7.0 [7.0, 7.1]	1.63 [1.59, 1.68]	99.4 [99.3, 99.5]
10.8	pitch	6.6 [6.5, 6.7]	7.0 [6.9, 7.0]	1.68 [1.63, 1.73]	99.4 [99.3, 99.5]
11.0	roll	8.2 [8.1, 8.3]	8.7 [8.6, 8.8]	1.73 [1.69, 1.77]	99.3 [99.1, 99.4]
13.1	pitch	6.6 [6.5, 6.7]	7.0 [6.9, 7.1]	1.74 [1.69, 1.79]	99.3 [99.2, 99.4]
13.5	roll	7.4 [7.3, 7.5]	7.8 [7.7, 7.9]	1.79 [1.74, 1.83]	99.2 [99.1, 99.3]
15.8	roll	7.9 [7.8, 8.0]	8.4 [8.3, 8.5]	1.79 [1.75, 1.84]	99.2 [99.1, 99.3]
15.8	pitch	7.9 [7.8, 8.0]	8.5 [8.4, 8.5]	1.79 [1.74, 1.84]	99.3 [99.2, 99.5]
17.8	roll	8.9 [8.8, 9.0]	9.4 [9.3, 9.5]	1.87 [1.81, 1.92]	99.1 [98.9, 99.3]
17.8	pitch	7.4 [7.3, 7.5]	8.0 [7.9, 8.1]	1.91 [1.85, 1.97]	99.2 [99.0, 99.4]
18.8	roll	6.5 [6.4, 6.6]	7.1 [7.0, 7.2]	1.88 [1.83, 1.93]	99.1 [98.9, 99.2]
20.1	pitch	7.5 [7.4, 7.6]	8.1 [8.0, 8.2]	2.02 [1.96, 2.09]	99.0 [98.8, 99.2]
21.9	roll	10.9 [10.8, 11.0]	11.4 [11.3, 11.5]	1.96 [1.91, 2.02]	98.9 [98.8, 99.1]
22.0	pitch	7.8 [7.7, 7.9]	8.1 [8.0, 8.2]	2.01 [1.94, 2.08]	99.1 [98.9, 99.2]
23.8	roll	9.9 [9.7, 10.0]	10.3 [10.2, 10.4]	2.05 [2.00, 2.10]	98.7 [98.5, 98.9]
24.1	pitch	8.8 [8.7, 8.9]	9.2 [9.1, 9.4]	2.11 [2.03, 2.18]	98.9 [98.7, 99.1]
25.5	roll	11.1 [11.0, 11.2]	11.5 [11.4, 11.6]	2.11 [2.06, 2.18]	98.4 [98.2, 98.7]
26.5	pitch	5.0 [4.9, 5.1]	5.4 [5.3, 5.6]	2.21 [2.14, 2.29]	98.8 [98.6, 99.0]
27.9	pitch	10.8 [10.6, 10.9]	11.4 [11.3, 11.6]	2.29 [2.22, 2.37]	98.8 [98.5, 99.0]

Table D.3: Results for Planar Characterisation Against Light Intensity. Per-trial values over n = 592-599 Frames.

Illuminance (lx)	Median depth error (mm) per-trial median [Q1, Q3]	Depth RMSE (mm) per-trial median [Q1, Q3]	Plane-fit RMS (mm) per-trial median [Q1, Q3]	Valid pixels (%) per-trial median [Q1, Q3]
0	4.80 [4.80, 5.20]	6.22 [5.97, 6.41]	1.80 [1.75, 1.84]	99.9 [99.9, 100.0]
17	7.50 [7.00, 7.50]	8.01 [7.76, 8.20]	2.74 [2.64, 2.82]	99.4 [99.3, 99.5]
65	5.20 [5.20, 5.60]	6.81 [6.67, 6.97]	2.84 [2.75, 2.94]	99.3 [99.2, 99.5]
78	7.50 [7.00, 7.50]	8.29 [8.04, 8.50]	3.44 [3.33, 3.56]	98.8 [98.5, 98.9]
91	7.00 [7.00, 7.50]	8.41 [8.17, 8.65]	4.14 [4.01, 4.29]	97.0 [96.7, 97.4]
151	7.50 [7.50, 7.90]	8.83 [8.60, 9.06]	4.43 [4.29, 4.59]	96.3 [95.9, 96.8]

D.2 Camera Configuration Results

Table D.4: Each value is over n = 15 scans. Fixed: Laser power = 150; Emitter = Laser; Fps = 30; Status = ok. Median taken over: Depth unit (3), Preset (5), Preset actual (5).

Resolution	Object	Accuracy RMSE (mm) median [Q1, Q3]	Signed bias (mm) median [Q1, Q3]	Completeness (full reference) at 2 mm (%) median [Q1, Q3]	F-score at 2 mm (%) median [Q1, Q3]
640 × 360	Big pyramid	2.14 [1.99, 2.19]	−0.013 [−0.019, −0.008]	50.7 [47.4, 52.7]	56.1 [52.9, 58.5]
	Big Egg Box	3.20 [2.86, 3.28]	−0.466 [−0.547, −0.396]	28.0 [25.5, 31.9]	33.9 [32.2, 37.6]
848 × 480	Big pyramid	1.70 [1.61, 1.77]	−0.001 [−0.007, 0.001]	65.7 [62.2, 68.5]	69.9 [67.2, 72.7]
	Big Egg Box	2.39 [2.34, 2.52]	−0.222 [−0.253, −0.115]	47.5 [45.0, 50.9]	52.3 [51.8, 55.4]
1280 × 720	Big pyramid	1.47 [1.34, 1.57]	−0.017 [−0.027, −0.002]	77.5 [69.6, 79.1]	79.9 [76.2, 81.0]
	Big Egg Box	2.07 [2.00, 2.21]	−0.100 [−0.118, −0.068]	57.1 [54.9, 61.4]	62.1 [59.8, 64.2]

Table D.5: Each value is over n = 15 scans. Fixed: Laser power = 150; Emitter = Laser; Fps = 30; Status = ok. Median taken over: Resolution (3), Preset (5), Preset actual (5).

Depth unit (mm)	Object	Accuracy RMSE (mm) median [Q1, Q3]	Signed bias (mm) median [Q1, Q3]	Completeness (full reference) at 2 mm (%) median [Q1, Q3]	F-score at 2 mm (%) median [Q1, Q3]
0.1	Big pyramid	1.62 [1.52, 2.03]	−0.016 [−0.021, −0.001]	64.6 [51.5, 70.3]	71.2 [57.3, 76.2]
	Big Egg Box	2.35 [2.26, 2.85]	−0.154 [−0.399, −0.113]	45.8 [31.6, 54.9]	52.9 [37.5, 58.7]
0.5	Big pyramid	1.70 [1.55, 1.85]	−0.006 [−0.012, 0.003]	62.2 [56.8, 75.2]	69.9 [63.2, 77.9]
	Big Egg Box	2.39 [2.15, 2.86]	−0.230 [−0.340, −0.101]	47.9 [31.6, 54.4]	52.6 [37.3, 61.2]
1	Big pyramid	1.74 [1.56, 2.01]	−0.008 [−0.023, −0.006]	62.9 [52.9, 73.4]	68.8 [58.8, 75.5]
	Big Egg Box	2.47 [2.23, 2.74]	−0.269 [−0.372, −0.104]	47.5 [35.8, 56.6]	51.8 [41.5, 60.8]

Table D.6: Each value is over n = 8–10 scans. Fixed: Laser power = 150; Emitter = Laser; Fps = 30; Status = ok. Median taken over: Resolution (3), Depth unit (3), Preset (5).

Preset actual	Object	Accuracy RMSE (mm) median [Q1, Q3]	Signed bias (mm) median [Q1, Q3]	Completeness (full reference) at 2 mm (%) median [Q1, Q3]	F-score at 2 mm (%) median [Q1, Q3]
Custom	Big pyramid	1.70 [1.56, 1.83]	−0.011 [−0.020, −0.001]	64.6 [56.4, 70.9]	69.9 [62.7, 75.2]
	Big Egg Box	2.43 [2.27, 2.71]	−0.232 [−0.383, −0.135]	48.0 [38.3, 53.5]	52.7 [43.8, 58.7]
Default	Big pyramid	1.88 [1.69, 2.14]	0.001 [−0.018, 0.005]	58.7 [48.2, 74.6]	63.2 [52.9, 75.1]
	Big Egg Box	2.51 [2.29, 2.99]	−0.353 [−0.388, −0.122]	46.7 [29.6, 56.3]	51.4 [35.2, 59.1]
High Accuracy	Big pyramid	1.74 [1.33, 2.14]	−0.017 [−0.019, −0.014]	59.9 [48.6, 62.2]	67.8 [54.8, 71.2]
	Big Egg Box	2.36 [2.00, 3.29]	−0.098 [−0.466, −0.060]	44.0 [25.1, 51.9]	51.8 [32.5, 60.1]
High Density	Big pyramid	1.60 [1.55, 1.86]	−0.007 [−0.014, −0.004]	72.3 [57.2, 78.8]	75.9 [63.6, 79.9]
	Big Egg Box	2.28 [2.05, 2.64]	−0.109 [−0.248, −0.104]	55.0 [36.2, 62.0]	59.2 [41.9, 64.4]
Medium density	Big pyramid	1.65 [1.48, 1.97]	−0.007 [−0.013, −0.001]	67.1 [53.6, 77.5]	71.2 [59.5, 80.2]
	Big Egg Box	2.59 [2.21, 3.20]	−0.236 [−0.493, −0.135]	47.1 [28.1, 55.0]	51.8 [33.9, 58.4]

D.3 Filter Configuration Results

Table D.7: Each value is over n = 4–5 scans. Fixed: Tilt = 0; Spatial filter on = Off; HDR merge on = Off; Status = ok. Median taken over: Rotation (4).

Object	Height (mm)	Configuration	Accuracy RMSE (mm) median [Q1, Q3]	Signed bias (mm) median [Q1, Q3]	Completeness (full reference at 2 mm (%) median [Q1, Q3
Big Egg Box	641	Baseline (no filters)	1.71 [1.67, 1.80]	−0.060 [−0.073, −0.053]	70.6 [68.0, 70.6
		Best overall	1.71 [1.70, 1.76]	−0.086 [−0.108, −0.086]	68.9 [68.9, 69.9
		Best for this shape & height	1.68 [1.66, 1.70]	−0.073 [−0.078, −0.072]	70.3 [69.8, 70.6
	701	Baseline (no filters)	1.99 [1.96, 1.99]	−0.047 [−0.060, −0.012]	63.5 [61.3, 64.0
		Best overall	1.88 [1.88, 1.89]	−0.055 [−0.064, −0.044]	62.6 [61.9, 62.8
		Best for this shape & height	1.89 [1.88, 1.91]	−0.041 [−0.089, −0.028]	11.2 [11.1, 11.2
	763	Baseline (no filters)	2.14 [2.12, 2.25]	−0.194 [−0.226, −0.174]	58.9 [57.0, 62.0
		Best overall	2.09 [2.08, 2.10]	−0.171 [−0.236, −0.154]	60.9 [60.2, 61.0
		Best for this shape & height	2.06 [2.03, 2.09]	−0.237 [−0.237, −0.226]	52.9 [52.6, 52.9
	810	Baseline (no filters)	2.15 [2.15, 2.35]	−0.103 [−0.150, −0.044]	54.6 [54.5, 56.7
		Best overall	2.11 [2.10, 2.14]	−0.085 [−0.090, −0.082]	56.1 [55.9, 56.9
		Best for this shape & height	2.09 [2.08, 2.12]	−0.110 [−0.115, −0.104]	46.7 [45.6, 47.0
	863	Baseline (no filters)	2.64 [2.55, 2.76]	−0.121 [−0.160, −0.016]	49.8 [48.7, 50.7
		Best overall	2.40 [2.37, 2.46]	−0.202 [−0.202, −0.201]	51.6 [50.1, 52.9
		Best for this shape & height	2.37 [2.27, 2.42]	−0.141 [−0.232, −0.136]	54.1 [49.1, 54.7
	933	Baseline (no filters)	2.95 [2.84, 3.20]	−0.345 [−0.385, −0.341]	43.3 [37.5, 45.2
		Best overall	2.80 [2.78, 2.86]	−0.311 [−0.362, −0.303]	43.6 [40.1, 44.6
		Best for this shape & height	2.89 [2.73, 2.96]	−0.221 [−0.322, −0.088]	47.6 [42.6, 52.5
Small Egg Box	641	Baseline (no filters)	2.04 [2.01, 2.05]	−0.109 [−0.113, −0.105]	54.2 [53.9, 55.7
		Best overall	2.04 [2.01, 2.06]	−0.112 [−0.113, −0.110]	54.7 [54.6, 55.3
		Best for this shape & height	2.00 [1.97, 2.01]	−0.107 [−0.110, −0.103]	55.3 [53.9, 55.3
	701	Baseline (no filters)	2.47 [2.43, 2.49]	−0.106 [−0.109, −0.053]	44.9 [42.9, 45.4
		Best overall	2.43 [2.41, 2.44]	−0.092 [−0.102, −0.088]	45.2 [45.2, 45.7
		Best for this shape & height	2.39 [2.39, 2.41]	−0.085 [−0.097, −0.076]	46.1 [45.4, 46.2
	763	Baseline (no filters)	2.67 [2.67, 2.68]	−0.194 [−0.199, −0.190]	36.5 [36.3, 37.7
		Best overall	2.63 [2.61, 2.63]	−0.182 [−0.183, −0.182]	36.5 [36.4, 36.7
		Best for this shape & height	2.59 [2.59, 2.60]	−0.180 [−0.195, −0.177]	37.3 [36.8, 37.5
	810	Baseline (no filters)	2.76 [2.76, 2.83]	−0.216 [−0.248, −0.216]	35.0 [34.3, 35.2
		Best overall	2.77 [2.77, 2.78]	−0.199 [−0.203, −0.195]	34.3 [33.9, 34.3
		Best for this shape & height	2.75 [2.74, 2.75]	−0.194 [−0.199, −0.175]	34.9 [34.1, 36.0
	863	Baseline (no filters)	3.07 [3.05, 3.20]	−0.208 [−0.213, −0.144]	32.3 [32.1, 32.4
		Best overall	2.99 [2.99, 3.01]	−0.163 [−0.186, −0.158]	32.7 [32.3, 33.0
		Best for this shape & height	2.98 [2.97, 2.99]	−0.184 [−0.191, −0.176]	25.3 [25.0, 25.5
	933	Baseline (no filters)	3.17 [3.14, 3.24]	−0.263 [−0.271, −0.259]	27.3 [26.5, 28.6
		Best overall	3.03 [3.02, 3.04]	−0.263 [−0.271, −0.260]	28.5 [28.0, 28.6
		Best for this shape & height	3.03 [3.03, 3.03]	−0.258 [−0.261, −0.257]	28.5 [28.3, 28.5

D.4 Fusion Combination Results

Table D.8: Each value is over n = 1 point clouds. Fixed: Status = ok.

Object	Height (mm)	View family	Accuracy RMSE (mm) median [Q1, Q3]	Signed bias (mm) median [Q1, Q3]	Completeness (full reference) at 2 mm (%) median [Q1, Q3]	F-score at 2 mm (%) median [Q1, Q3]
Big Egg Box	641	all	1.15	−0.054	88.8	90.9
		single	1.86	−0.096	65.3	68.7
	701	all	1.34	−0.061	81.0	83.6
		single	1.94	−0.041	64.7	67.8
	763	all	1.57	−0.087	74.0	76.9
		single	2.33	−0.205	59.2	63.0
	810	all	1.76	−0.108	66.6	70.1
		single	2.11	−0.057	60.4	64.4
	863	all	1.89	−0.098	62.0	65.3
		single	2.24	−0.219	53.1	57.5
	933	all	2.31	−0.168	51.9	55.0
		single	3.20	−0.362	31.8	36.5
Small Egg Box	641	all	1.93	−0.082	54.4	60.8
		single	2.09	−0.081	53.8	59.8
	701	all	2.15	−0.083	48.9	55.4
		single	2.56	−0.127	40.4	46.8
	763	all	2.43	−0.072	43.5	50.3
		single	2.71	−0.192	34.9	42.0
	810	all	2.63	−0.113	39.1	46.0
		single	2.73	−0.189	34.4	41.9
	863	all	2.72	−0.080	38.6	45.4
		single	3.12	−0.153	31.5	39.1
	933	all	2.94	−0.105	35.1	41.7
		single	3.09	−0.253	28.4	35.4

Appendix E

Characterisation Plots

E.1 Filter Heatmaps

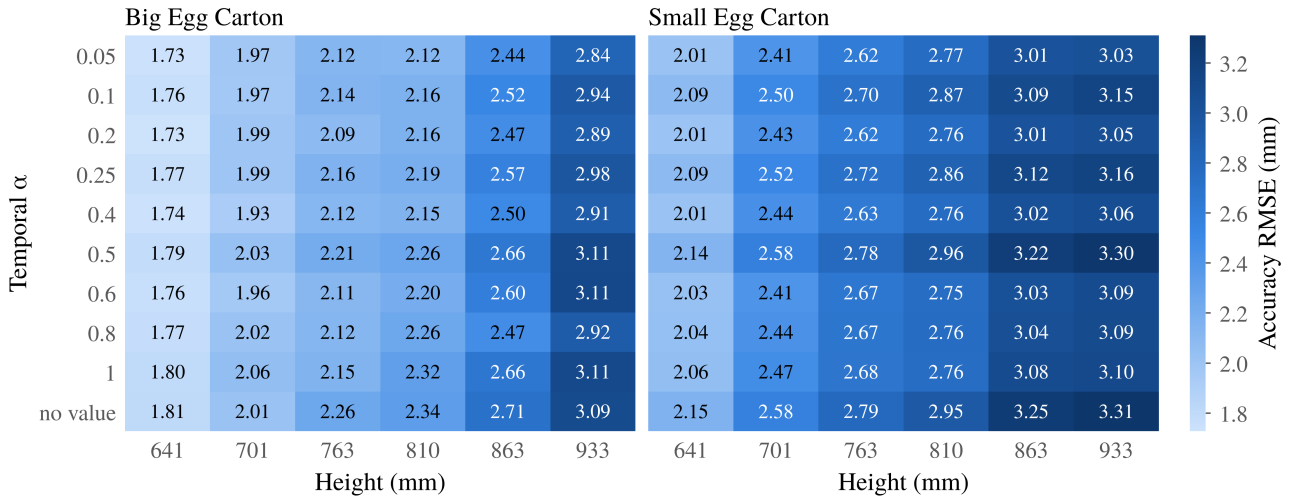


Figure E.1: Median with IRQ % CI of the mean of n = 5–295 scans per cell. Fixed: Tilt = 0. Averaged over: Rotation (4), Decimation on (2), Decimation magnitude (7), Spatial filter on (2), Spatial magnitude (6), Spatial α (8) and 8 more. Spanning 257 filter configurations. 2,460 rows without a value excluded.

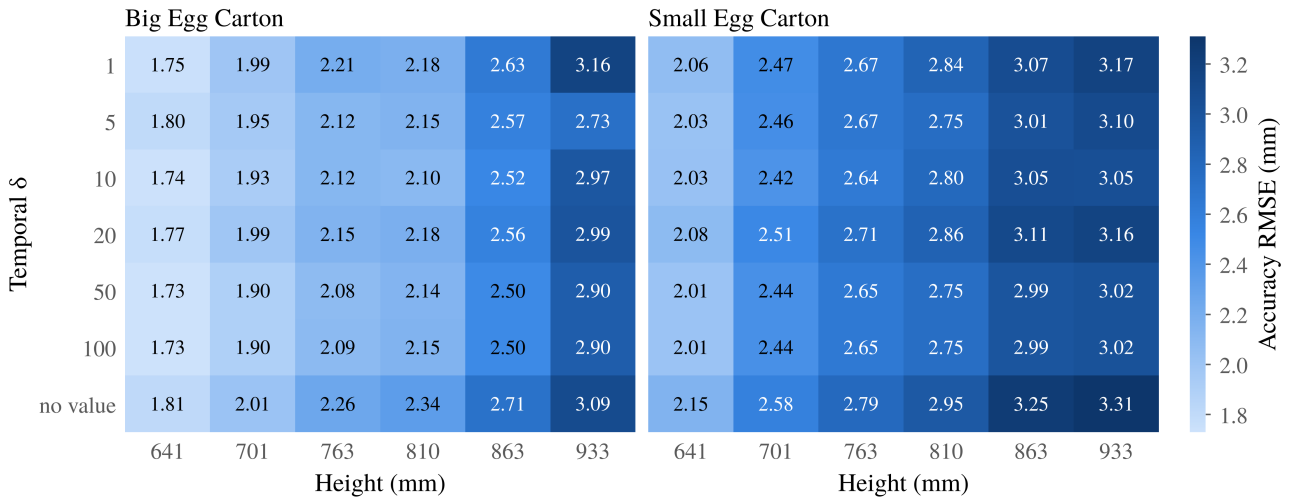


Figure E.2: Median with IRQ % CI of the mean of n = 5–815 scans per cell. Fixed: Tilt = 0. Averaged over: Rotation (4), Decimation on (2), Decimation magnitude (7), Spatial filter on (2), Spatial magnitude (6), Spatial α (8) and 8 more. Spanning 257 filter configurations. 2,460 rows without a value excluded.

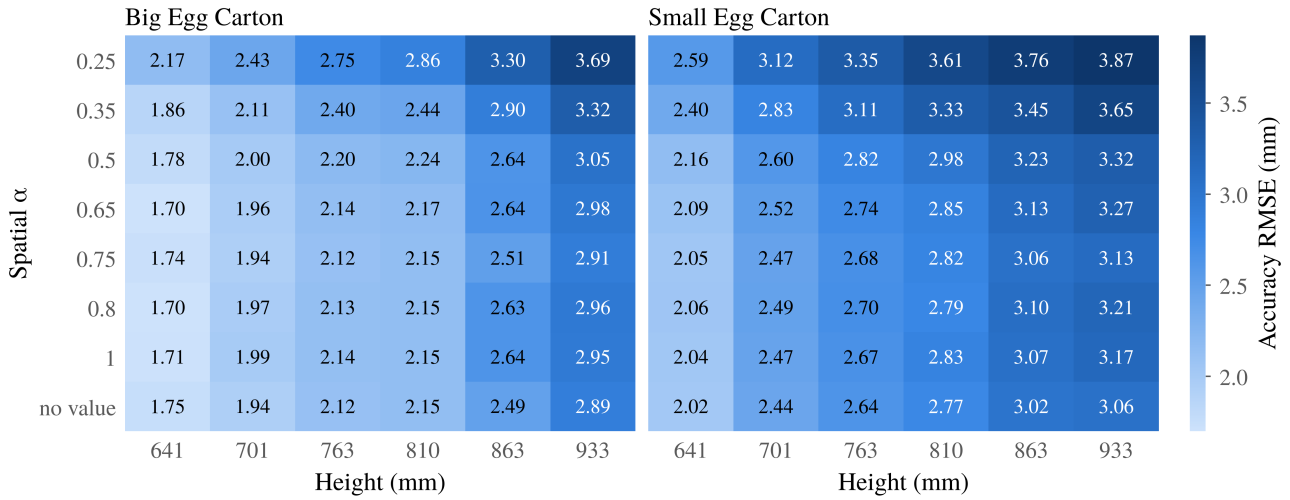


Figure E.3: Median with IQR of n = 5–315 scans per cell. Fixed: Tilt = 0. Median taken over: Rotation (4), Decimation on (2), Decimation magnitude (7), Spatial magnitude (6), Spatial δ (9), Spatial hole fill (7) and 8 more. Spanning 257 filter configurations. 2,460 rows without a value excluded.

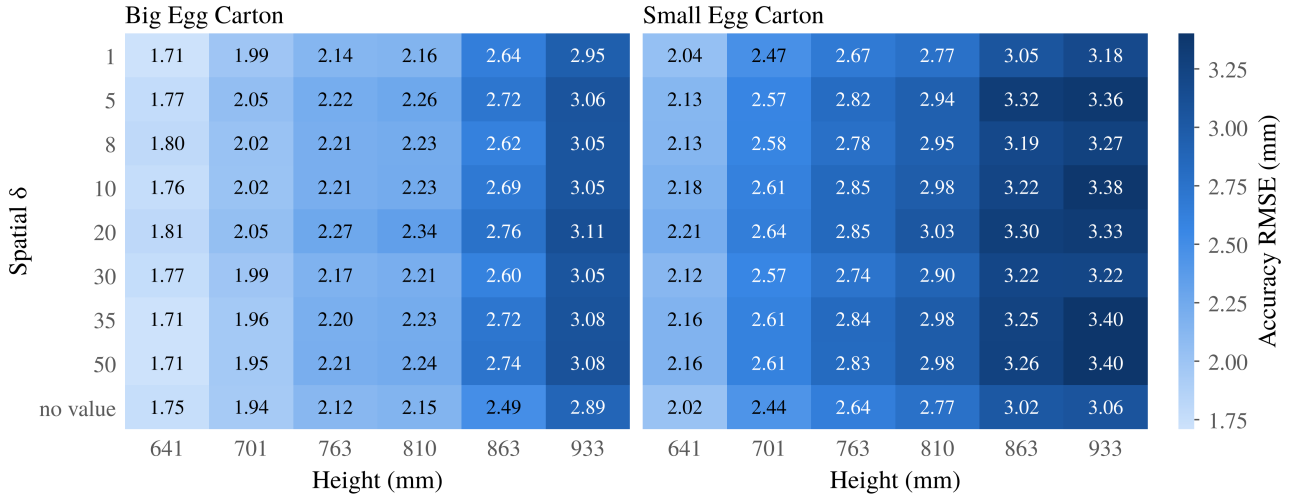


Figure E.4: Median with IQR of n = 5–385 scans per cell. Fixed: Tilt = 0. Median taken over: Rotation (4), Decimation on (2), Decimation magnitude (7), Spatial magnitude (6), Spatial α (8), Spatial hole fill (7) and 8 more. Spanning 257 filter configurations. 2,460 rows without a value excluded.

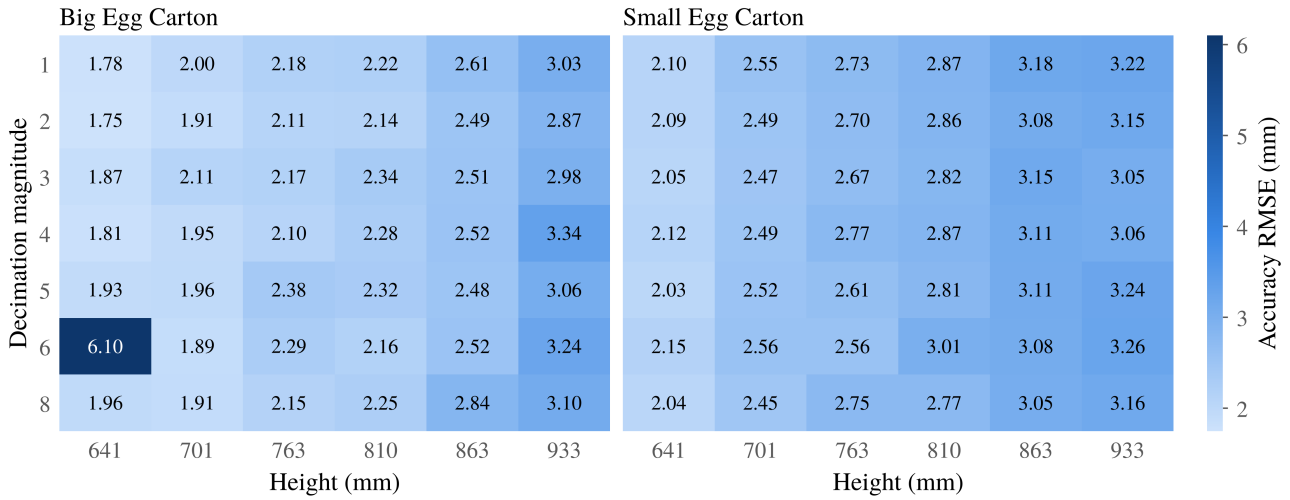


Figure E.5: Median with IQR of n = 5–950 scans per cell. Fixed: Tilt = 0. Median taken over: Rotation (4), Spatial filter on (2), Spatial magnitude (6), Spatial α (8), Spatial δ (9), Spatial hole fill (7) and 8 more. Spanning 257 filter configurations. 2,460 rows without a value excluded.

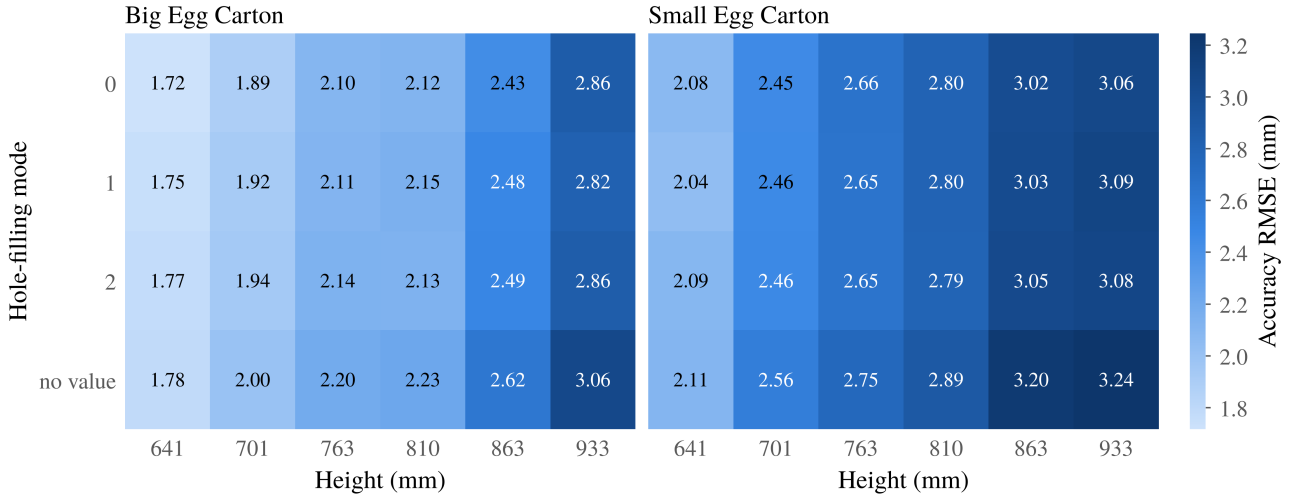


Figure E.6: Median with IQR of n = 55–915 scans per cell. Fixed: Tilt = 0. Median taken over: Rotation (4), Decimation on (2), Decimation magnitude (7), Spatial filter on (2), Spatial magnitude (6), Spatial α (8) and 8 more. Spanning 257 filter configurations. 2,460 rows without a value excluded.

E.2 Combination Heatmaps

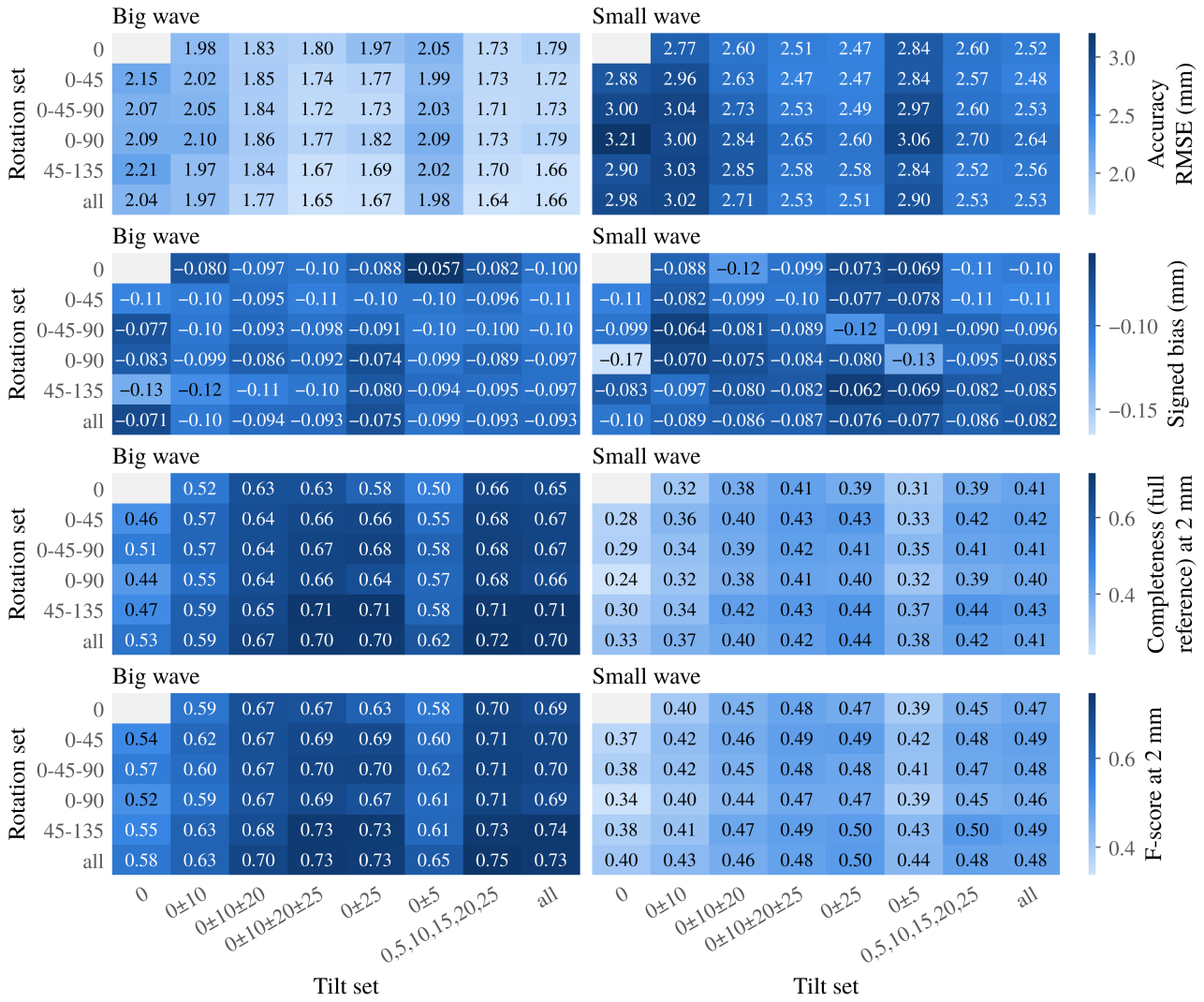
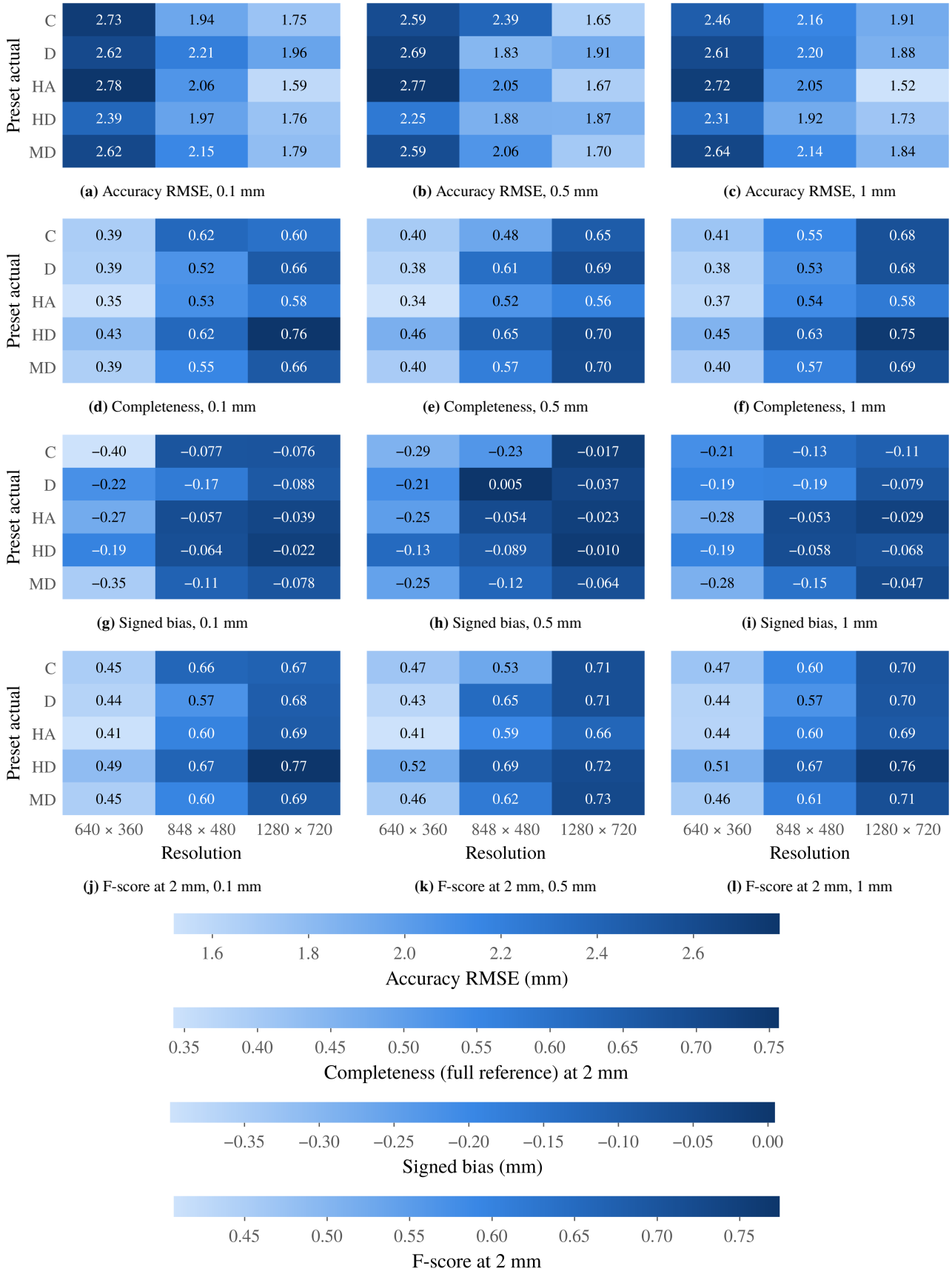


Figure E.7: Grey cells: no data for that combination. Median with IQR of n = 6 point clouds per cell. Fixed: Kind = fused; Fusion cell = 0.0005; Scan voxel = 0.0005; Status = ok. Median taken over: Height (6).

E.3 Parameter Heatmap



(a) Accuracy RMSE, 0.1 mm **(b)** Accuracy RMSE, 0.5 mm **(c)** Accuracy RMSE, 1 mm

(d) Completeness, 0.1 mm **(e)** Completeness, 0.5 mm **(f)** Completeness, 1 mm

(g) Signed bias, 0.1 mm **(h)** Signed bias, 0.5 mm **(i)** Signed bias, 1 mm

(j) F-score at 2 mm, 0.1 mm **(k)** F-score at 2 mm, 0.5 mm **(l)** F-score at 2 mm, 1 mm

Figure E.8: Median with IQR of n = 1–3 scans per cell. Fixed: Laser power = 150; Emitter = Laser; Fps = 30; Status = ok. Median taken over: Object (2), Preset (5).

Appendix F

Validation Results

F.1 Reference Shapes Rugosity

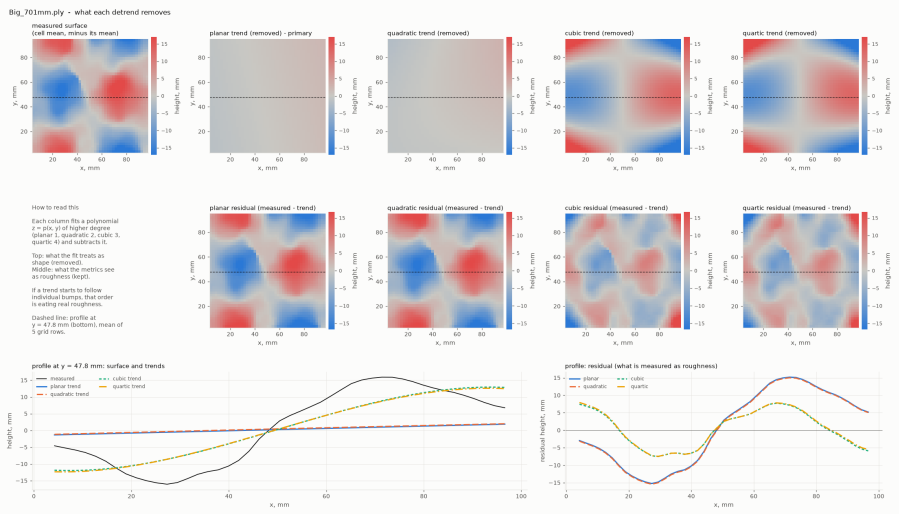


Figure F.1: Big Wave Detrend Explained Single

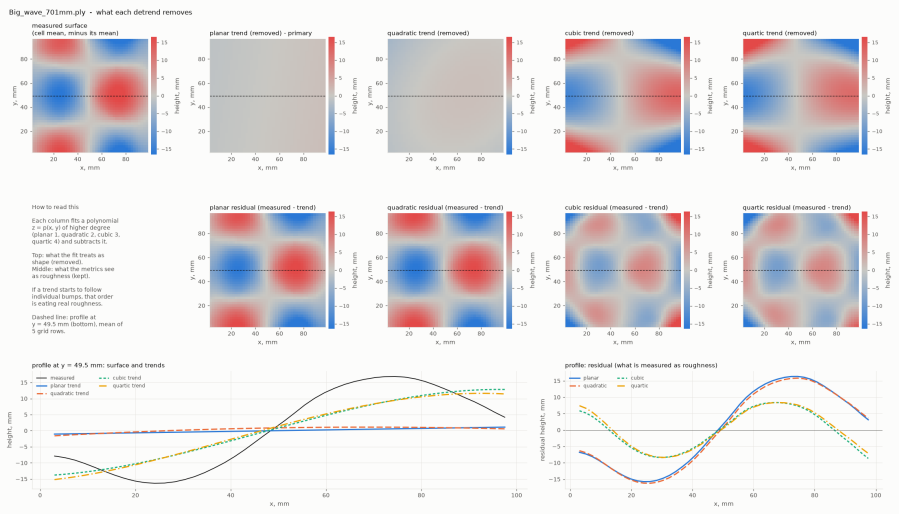


Figure F.2: Big Wave Detrend Explained Fused

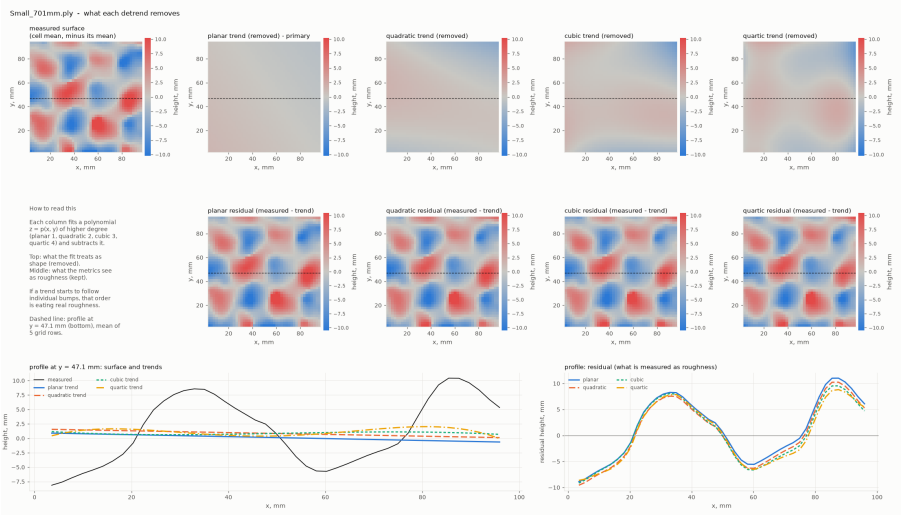


Figure F.3: Small Wave Detrend Explained Single

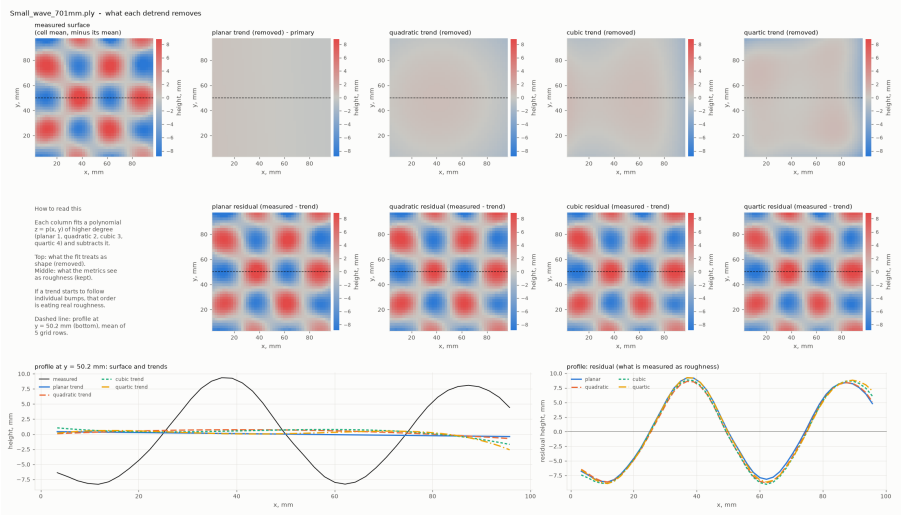


Figure F.4: Small Wave Detrend Explained Fused

Appendix B: Ethics approval

Appendix C: Topic description

The original topic description used during the project selection process is included below. This topic has since been further refined, and the exact problem that was investigated is described in Section 1.2.

Appendix D: ECSA Graduate Attributes

Project mapping to ECSA Graduate Attributes

The table below provides a structured reflection on where each of the required Graduate Attributes (GAs) as defined by Engineering Council of South Africa (ECSA) associated with the final year project is met. This includes reference to the specific Chapter/Section (where applicable).

GA	Graduate Attribute Description	Student reflection: How this outcome was achieved during the project
GA1	*Problem solving*	
GA4	*Investigations, experiments and data analysis*	
GA5	*Use of engineering tools*	
GA6	*Professional and technical communication*	
GA8	*Individual and collaborative teamwork*	
GA9	*Independent learning ability*	

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